

Full waveform numerical simulations of seismoelectromagnetic wave conversions in fluid-saturated stratified porous media

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[1] We present a full-waveform modeling technique of the coupled seismoelectromagnetic wave propagation in fluid-saturated stratified porous media. Our simulation code uses the macroscopic governing equations derived by *Pride* [1994], which couple Biot's theory and Maxwell equations via flux/force transport equations. In this theory the coupling mechanism is explained by electrokinetic effects taking place at the pore level. The synthetic seismoelectrograms and seismomagnetograms are computed by extending the generalized reflection and transmission matrix method and by using a discrete wave number integration of the global reflectivity obtained in the frequency wave number domain. Synthetic time sections and snapshots of the wave propagation are used to study the seismic, electromagnetic, and seismoelectromagnetic waves properties in fluid-saturated layered porous media. Two wave phenomena are investigated: (1) the electric and magnetic fields induced by the propagation of a seismic perturbation in a homogeneous porous medium and (2) the electromagnetic waves generated at depth when seismic waves propagate through a vertically heterogeneous porous medium. Concentrating on the second effect, we show that the zone which effectively contributes to the generation of EM disturbances along a plane interface coincides with the first Fresnel zone associated with a seismic-to-electromagnetic wave conversion. A numerical sensitivity study shows that the EM waves generated at depth by the passage of seismic waves through an interface are particularly sensitive to contrasts in porosity, permeability, fluid salinity, and fluid viscosity. Our numerical simulations highlight the potential of artificially generated seismoelectromagnetic converted waves for the characterization of the subsurface and its fluid content.

INDEX TERMS: 0902 Exploration Geophysics: Computational methods, seismic; 0689 Electromagnetics: Wave propagation (4275); 3210 Mathematical Geophysics: Modeling; 5139 Physical Properties of Rocks: Transport properties; **KEYWORDS:** modeling, porous media, electrokinetic coupling, seismo-electromagnetic waves, exploration geophysics

1. Introduction

[2] Several mechanisms are likely to generate couplings between seismic and electromagnetic (EM) energy in the subsurface. A review of the main effects of interest to geophysicists, namely, electrokinetic and piezoelectric phenomena and variations in electrical resistivity, is given by *Russel et al.* [1997]. The above mentioned effects were mainly studied as potential geophysical exploration tools. They were also investigated to analyze and interpret various EM disturbances sometimes observed before or during an earthquake [*Ishido and Mizutani*, 1981; *Park et al.*, 1993]. Over the past decades, several field experiments were conducted to understand the physics of electrokinetic phenomena within the subsurface from active seismoelectric measurements [*Thompson*, 1936; *Martner and Sparks*,

1959; *Broding et al.*, 1963; *Long and Rivers*, 1975]. Although electrokinetic effects were discovered a long time ago, successful field experiments performed in recent years [*Thompson and Gist*, 1993] have stimulated new interest in this particular mechanism [*Butler et al.*, 1996; *Mikhailov et al.*, 1997, 2000; *Beamish*, 1999; *Garambois and Dietrich*, 2001]. This interest has further increased with the work of *Pride* [1994], who derived macroscopic equations controlling the coupled electromagnetic and seismic wave fields in a fluid-saturated porous medium. These analytical developments opened the possibility to numerically simulate the electrokinetic coupling phenomena for a better understanding of their properties and their potential use for the detection of fluids in the subsurface.

[3] It has been theoretically shown that when a seismic disturbance propagates through a fluid-saturated porous medium, electrokinetic coupling phenomena generate EM disturbances of two different kinds [*Pride and Haartsen*, 1996]. Owing to the presence of an electric double layer near the contact surface separating a mineral and a fluid electrolyte, a local macroscopic electric field (magnetic field, respectively) is created in a homogeneous medium,

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which accompanies the seismic P wave (S wave, respectively) within the support of the seismic signal. These fields have no extent outside of the seismic wave. Such a mechanism can explain many observations made on seismoelectric conversions [Thompson, 1936; Broding *et al.*, 1963; Butler *et al.*, 1996; Mikhailov *et al.*, 1997; Beamish, 1999]. It has been shown experimentally [Garambois and Dietrich, 2001] that the electric fields generated by the passage of seismic waves are consistent in their waveform and amplitude with those predicted by the theoretical equations [Pride and Haartsen, 1996]. Garambois and Dietrich [2001] also derived analytical expressions for the transfer functions between the seismic signals and the corresponding electric or magnetic fields in the low-frequency range. In the same study, they showed that the electric fields propagating within P waves are particularly sensitive to the electric properties of the saturating fluid (salinity and dielectric permittivity) and that the magnetic fields propagating within S waves strongly depend on the viscosity of the fluid saturating the porous medium. A similar investigation was made by Mikhailov *et al.*, [2000] with Stoneley waves in a borehole.

[4] The second effect predicted by Pride and Haartsen [1996] occurs when a mechanical disturbance propagates through a heterogeneous medium. When crossing an interface between two porous layers having different properties, a seismic wave generates a time-varying charge separation which acts as a dipole radiating electromagnetic energy independently of the seismic wave. The existence of this second-order effect was numerically confirmed by Haartsen and Pride [1997], who showed that it can explain the observations made by Thompson and Gist [1993], notably the fact that EM signals may be created at contrasts in mechanical or electric properties. Accordingly, it might be possible to detect and characterize the contrasts in fluid properties at depth with the resolution of seismic methods. Unfortunately, not many experiments show clear seismic-to-EM wave conversions at depth. The main difficulties arise from the high level of background EM noise in nature, the usually high amplitudes of the first effect discussed previously, and the low amplitude of the EM disturbance generated at an interface. Nevertheless, Martner and Sparks [1959] showed clear EM precursory pulses generated at the base of the weathered layer by the passage of seismic waves. Garambois and Dietrich [2001] also recorded EM disturbances originating from the top of a water table with an amplitude pattern characteristic of a dipole radiation.

[5] Even though conversions between seismic and EM energy are now better understood, a numerical modeling approach remains necessary to quantify these effects and to design an optimum acquisition geometry in the field. Conversely, the detailed comparison between numerical simulations and real data will be useful to refine the theory which still lacks extensive experimental checking. Using their previous work [Pride, 1994; Pride and Haartsen, 1996], Haartsen and Pride [1997] developed a simulation program for the coupled seismic and EM wave propagation in layered porous media. Their computation technique uses the general matrix method, which was originally designed for seismic wave propagation in elastic layered media by Chin *et al.* [1984]. Haartsen and Pride considerably simplified the problem by deriving the seismo-EM eigen

responses for plane waves, by considering various excitation sources, and by expressing the free surface boundary conditions. Their numerical applications illustrate well the seismic-to-EM wave coupling phenomena with simple examples.

[6] In this paper, we extend Haartsen and Pride's [1997] work with a slightly different modeling code to address some important points such as the sensitivity of the EM waves to the properties of the porous material. In our approach the generalized reflection and transmission matrix (GRTM) method [Kennett and Kerry, 1979] is adapted to handle the coupled seismic and electromagnetic wave propagation in a fluid-saturated stratified porous medium. This modeling technique presents unique advantages compared to propagator matrix solutions, such as the possibility to easily compute partial solutions to the complete response. This property is especially useful when analyzing signals which differ by several orders of magnitude since it is then possible to separately compute the individual contributions to the total wave field. Alternatively, the small amplitude (and usually hidden) contributions can also be artificially enhanced by modifying the reflection and transmission coefficients in the GRTM method.

[7] The simulation program is constructed by retaining the general structure of existing codes designed for acoustic or elastic wave propagation in layered media (e.g., Dietrich, [1988]). One of the particular features of the coupled seismic and EM wave propagation is the coexistence of waves which propagate at radically different velocities. As we will see, this peculiarity can easily be dealt with. Considering that wave propagation in stratified media is a well-developed subject, we will restrict ourselves to the presentation of the theoretical background and to the discussion of some particular points which differ from elastic wave modeling.

[8] As a first test of our program, we show that the general reciprocity theorem of Green's tensor is satisfied for seismic, electromagnetic and seismoelectromagnetic waves. We then present snapshots of seismic displacements and electric and magnetic fields for a three-layer model that illustrate the two seismo-EM coupling effects. In the second part of the paper, we concentrate on the seismic-to-EM wave conversions at depth to determine the region where these disturbances are generated, by using the concept of Fresnel zones. We also estimate the expected amplitudes and radiation patterns of the converted EM waves and present a sensitivity study to determine the most influent parameters.

2. Mathematical Formulation

[9] Pride [1994] derived, in the frequency domain, the macroscopic dynamic equations controlling the coupled seismic and electromagnetic wave propagation in a fluid-saturated porous medium. In the particular case of stratified media, Haartsen and Pride [1997] transformed this set of equations into the familiar system of first-order ordinary depth-dependent differential equations written in the form

$$\frac{\partial}{\partial z} \mathbf{B} = \mathbf{A} \mathbf{B} + \mathbf{F}, \quad (1)$$

where $\mathbf{B}$ denotes a displacement-stress-EM vector and $\mathbf{F}$ is the source vector. Matrix $\mathbf{A}$ is defined below. The transformations used to obtain equation (1) are similar to the ones originally developed by *Kennett and Kerry* [1979] in the elastic case, i.e., a cylindrical wave decomposition in the frequency-wave number domain followed by decompositions of each wave field into vertical and horizontal components, as first suggested by *Hudson* [1969].

[10] The set of equations thus obtained can then be split into two independent systems in a homogeneous and isotropic whole space. The so-called *PSVTM* set couples the propagation of fast P waves (P_f), Biot slow waves (P_s), S waves polarized in the vertical plane of propagation (SV), and EM waves with a transverse polarization of the magnetic field (TM). The second set, referred to as *SHTE*, couples the propagation of horizontally polarized S waves (SH) with EM waves characterized by a transverse polarization of the electric field (TE).

[11] The $\mathbf{B}$ vectors of the two systems are written as

$$\mathbf{B}^{PSVTM} = [\hat{u}_1, \hat{u}_z, \hat{w}_z, \hat{\tau}_1, \hat{\tau}_{zz}, -\hat{P}, \hat{H}_2, \hat{E}_1] \quad (2)$$

$$\mathbf{B}^{SHTE} = [\hat{u}_2, \hat{\tau}_2, \hat{H}_1, \hat{E}_2], \quad (3)$$

where $\hat{\mathbf{u}}$ is the macroscopic average displacement of the solid grains, $\hat{\mathbf{w}}$ is the Darcy displacement defined as $\hat{\mathbf{w}} = \phi(\hat{\mathbf{u}}_f - \hat{\mathbf{u}})$, where ϕ is porosity and $\hat{\mathbf{u}}_f$ is the macroscopic average displacement of the fluid, $\hat{\boldsymbol{\tau}}$ is the bulk stress tensor defined as $\hat{\boldsymbol{\tau}} = \phi\hat{\boldsymbol{\tau}}_f + (1 - \phi)\hat{\boldsymbol{\tau}}_s$, where $\hat{\boldsymbol{\tau}}_s$ and $\hat{\boldsymbol{\tau}}_f$ are the average stress tensors in the grains and fluid, respectively. $\hat{P}$ is the fluid pressure, while $\hat{H}$ and $\hat{E}$ represent the magnetic and electric fields, respectively. The boundary conditions [*Pride and Haartsen*, 1996] require the continuity of vector $\mathbf{B}$ across an interface.

2.1. Decomposition of the Seismo-EM Wave Fields

[12] In a homogeneous fluid-saturated porous layer the various wave fields, i.e., displacements, stresses, and electromagnetic fields, can be split up into upgoing and downgoing wave contributions. This permits to connect the displacement-stress-EM vector $\mathbf{B}$ to a wave vector $\mathbf{V}$ containing the upward and downward potentials (complex amplitudes) of all wave types, that is,

$$\mathbf{B} = \mathbf{D}\mathbf{V}, \quad (4)$$

where

$$\mathbf{V}^{PSVTM} = [V_U^{P_f}, V_U^{P_s}, V_U^{SV}, V_U^{TM}, V_D^{P_f}, V_D^{P_s}, V_D^{SV}, V_D^{TM}] \quad (5)$$

$$\mathbf{V}^{SHTE} = [V_U^{SH}, V_U^{TE}, V_D^{SH}, V_D^{TE}], \quad (6)$$

and where the subscripts U and D stand for “upward” and “downward”, respectively.

[13] The columns of matrix $\mathbf{D}$ contain the eigenvectors of matrix $\mathbf{A}$ defined in equation (1). The elements of $\mathbf{A}$ are derived by *Haartsen and Pride* [1997] for an isotropic and homogeneous whole space for both *PSVTM* and *SHTE* systems. As there is a free choice to scale each eigenvector,

we follow *Pride and Haartsen* [1996] and normalize the eigenvectors with respect to the time average of the Poynting vector across a horizontal plane. Such a normalization guarantees the consistency between poroelastic and electromagnetic energy transfers across an interface. The sizes of matrix $\mathbf{D}$ in the *PSVTM* and *SHTE* cases are (8×8) and (4×4) , respectively.

[14] By analogy with the formulation used by *Kennett and Kerry* [1979], we can now partition matrix $\mathbf{D}$ in the form

$$\mathbf{D} = \begin{pmatrix} \mathbf{M}_U & \mathbf{M}_D \\ \mathbf{N}_U & \mathbf{N}_D \\ \mathbf{Mf}_U & \mathbf{Mf}_D \\ \mathbf{Ef}_U & \mathbf{Ef}_D \end{pmatrix}, \quad (7)$$

where $\mathbf{M}_U$ and $\mathbf{M}_D$ transform the upgoing and downgoing wave potentials into displacements and $\mathbf{N}_U$ and $\mathbf{N}_D$ do the same for the stresses. For the *PSVTM* wave system these four operators are (3×4) submatrices. $\mathbf{Mf}_U$ and $\mathbf{Mf}_D$ are the magnetic field transformations; $\mathbf{Ef}_U$ and $\mathbf{Ef}_D$ are the electric field transformations. These quantities are (1×4) matrices for the *PSVTM* system. All of the above mentioned matrices reduce to (1×2) matrices for the *SHTE* wave system.

2.2. Reflection and Transmission at an Interface

[15] Equations (5) and (6) indicate that an incident wave at an interface separating two porous media generates four reflected and four transmitted waves in the *PSVTM* system, and two reflected and two transmitted waves in the *SHTE* system. In this section, we follow the method described by *Kennett and Kerry* [1979] to obtain the reflection and transmission coefficients at an interface.

[16] The boundary conditions, which set the continuity of displacements, stresses, and electromagnetic fields at an interface located at depth z , imply that

$$\mathbf{B}(z^-) = \mathbf{B}(z^+). \quad (8)$$

Thus, from equation (4) we can introduce the wave propagator matrix $\mathbf{Q}$

$$\mathbf{V}_-(z^-) = \mathbf{Q}(z^-, z^+) \mathbf{V}_+(z^+) \quad (9)$$

with

$$\mathbf{Q}(z^-, z^+) = \mathbf{D}_-^{-1}(z) \mathbf{D}_+(z^+). \quad (10)$$

$\mathbf{Q}$ is a (8×8) matrix for the *PSVTM* system and a (4×4) matrix in the *SHTE* case. In our implementation the inverse of $\mathbf{D}$ is computed numerically, although it should be possible to calculate it analytically by using the symmetry properties of $\mathbf{A}$, as first suggested by *Haartsen and Pride* [1997].

[17] We now consider a partitioning of $\mathbf{Q}$, so that

$$\mathbf{Q}(z^-, z^+) = \begin{pmatrix} \mathbf{Q}_{UU} & \mathbf{Q}_{UD} \\ \mathbf{Q}_{DU} & \mathbf{Q}_{DD} \end{pmatrix}, \quad (11)$$

where $\mathbf{Q}_{UU}$, $\mathbf{Q}_{UD}$, $\mathbf{Q}_{DU}$, and $\mathbf{Q}_{DD}$ are (4×4) matrices in the *PSVTM* wave system and (2×2) matrices in the *SHTE* wave system. With this formulation the reflection

and transmission matrices at an interface are directly obtained from the equations derived by *Kennett and Kerry* [1979]:

$$\mathbf{T}_D = \mathbf{Q}_{DD}^{-1} \quad (12)$$

$$\mathbf{R}_D = \mathbf{Q}_{UD} \mathbf{Q}_{DD}^{-1} \quad (13)$$

$$\mathbf{R}_U = -\mathbf{Q}_{DD}^{-1} \mathbf{Q}_{DU} \quad (14)$$

$$\mathbf{T}_U = \mathbf{Q}_{UU} - \mathbf{Q}_{UD} \mathbf{Q}_{DD}^{-1} \mathbf{Q}_{DU}. \quad (15)$$

In the *PSVTM* wave system the reflection and transmission matrices are (4×4) matrices which contain the reflection and transmission coefficients of each wave type at an interface. For example, the $\mathbf{R}_D$ matrix is written in the form

$$\mathbf{R}_D = \begin{pmatrix} r_D^{P_f P_f} & r_D^{P_f P_s} & r_D^{P_f SV} & r_D^{P_f TM} \\ r_D^{P_s P_f} & r_D^{P_s P_s} & r_D^{P_s SV} & r_D^{P_s TM} \\ r_D^{SV P_f} & r_D^{SV P_s} & r_D^{SV SV} & r_D^{SV TM} \\ r_D^{TM P_f} & r_D^{TM P_s} & r_D^{TM SV} & r_D^{TM TM} \end{pmatrix}. \quad (16)$$

The term $r_D^{TMP_f}$ represents the amplitude of an upwardly propagating *TM* wave generated from the conversion of a unit amplitude downwardly incident seismic wave P_f at a plane interface. The reflection and transmission matrices are (2×2) matrices in the *SHTE* system.

[18] The specific boundary conditions for the coupled seismic and EM wave propagation at the free surface have been expressed by *Haartsen and Pride* [1997]. These boundary conditions require that (1) the normal and tangential stresses and normal fluid pressure vanish and (2) the electromagnetic fields obey the regular boundary conditions at the interface separating the near-surface porous medium and the air. From these relationships, *Haartsen and Pride* [1997] derived the reflection matrices $\mathbf{R}_U^f$ at the free surface for the *PSVTM* and *SHTE* wave systems.

2.3. Generalized Reflection and Transmission Matrix Method

[19] The generalized reflection and transmission matrix method [*Kennett and Kerry*, 1979] allows us to express the full seismic and electromagnetic response of a fluid-saturated stratified poroelastic half-space as a combination of (4×4) matrices for the *PSVTM* wave system and (2×2) matrices for the *SHTE* wave system.

[20] For a receiver above the source the solution is written as

$$\mathbf{W} = (\mathbf{C}_U^R + \mathbf{C}_D^R \mathbf{R}_U^{FR}) [\mathbf{I} - \mathbf{R}_D^{RS} \mathbf{R}_U^{FR}]^{-1} \cdot \mathbf{T}_U^{RS} [\mathbf{I} - \mathbf{R}_D^{SL} \mathbf{R}_U^{FS}]^{-1} (\mathbf{R}_D^{SL} \Sigma_D - \Sigma_U), \quad (17)$$

where $\mathbf{W}$ represents a subset of wave fields among the various wave fields composing the displacement-stress-EM vector $\mathbf{B}$. In practice, we compute the grain displacements and the electromagnetic fields at the receiver, i.e., $\mathbf{W} = (\hat{u}_1, \hat{u}_2, \hat{H}_2, \hat{E}_1)$ in the *PSVTM* case and $\mathbf{W} = \hat{u}_2, \hat{E}_2$ in the *SHTE* case. Matrices $\mathbf{R}_U^{FR}$, $\mathbf{R}_D^{RS}$, $\mathbf{R}_U^{FR}$, $\mathbf{R}_D^{SL}$, and $\mathbf{T}_U^{RS}$ denote the

overall reflection and transmission matrices in various parts of the medium. Following the notations of *Kennett and Kerry* [1979], F denotes the free surface, R denotes the receiver level, S denotes the source level, and L denotes the lower half-space. The generalized reflection and transmission matrices $\mathbf{R}_U^{FR}$, $\mathbf{R}_D^{RS}$, $\mathbf{R}_U^{FR}$, $\mathbf{R}_D^{SL}$, and $\mathbf{T}_U^{RS}$ are recursively constructed for a stack of uniform layers from the reflection and transmission matrices at an interface expressed previously [*Kennett and Kerry*, 1979]. $\mathbf{C}_U^R$ and $\mathbf{C}_D^R$ are matrices converting the upgoing and downgoing wave amplitudes into the desired wave fields at the receiver. They are expressed from subpartitions of matrix $\mathbf{D}$ (equation (7)). Finally, $\mathbf{R}_D^{SL} \Sigma_D - \Sigma_U$ represents the total contribution of upgoing waves generated by the excitation source at the source level. The various source terms considered in this study are given in the Appendix. The computation of equation (17) must be repeated for each plane wave component, i.e., for each frequency and each horizontal wave number considered.

[21] Equation (17) can be read from right to left by distinguishing three main contributions: the source term, the global reflectivity of the stratified medium, and, finally, the conversions of the wave potentials into displacements and EM fields at the receiver. Since the total response is expressed in terms of reflection and transmission matrices, it is possible, in the GRTM approach, to remove internal reverberations as well as multiple reflections in various parts of the layering. For the purpose of studying seismic-to-EM conversions at depth the ability to modify or cancel specific reflection or transmission coefficients is also very important. Indeed, the weakness of the converted EM waves compared to the electric or magnetic fields that propagate along with seismic waves can be overcome by scaling the corresponding reflection or transmission coefficients ($r_D^{TMP_f}$, $r_U^{TMP_f}$, for example) in order to artificially enhance these contributions in graphic displays.

[22] Let us now consider the source terms Σ_U and Σ_D . The source $\mathbf{S}$, with azimuthal order m at depth $z = z_s$ generates a discontinuity in the wave vector $\mathbf{V}$ across the horizontal plane containing the source, which is written as

$$[\mathbf{V}(z_s)]_+^+ = \mathbf{V}(z_s^+) - \mathbf{V}(z_s^-) = \Sigma(z_s) = \mathbf{D}^{-1}(z_s) \mathbf{S}(m, z_s). \quad (18)$$

The source potentials can therefore be partitioned into upward and downward contributions:

$$\Sigma(z_s) = \begin{pmatrix} \Sigma_U(z_s) \\ \Sigma_D(z_s) \end{pmatrix}, \quad (19)$$

where Σ_U and Σ_D are defined from the upgoing and downgoing wave potentials. We consider here both mechanical and electric point source excitations. Source vectors associated with various source types were given by *Haartsen and Pride* [1997]. We reproduce some of their analytical results in Appendix A and complete them with additional expressions required to check the reciprocity of the Green's tensor addressed in section 3.

2.4. Back in the Time-Distance Domain

[23] The displacements, stresses, and electromagnetic wave fields have so far been obtained in the frequency wave number domain (ω, k) for a given type of excitation.

These fields can be transformed into the time-distance domain by using Fourier and Hankel transforms of the plane wave responses. In a cylindrical coordinate system (r, θ, z) the displacements, stresses, and EM wave fields are written as a double integral over the angular frequencies ω and wave numbers k and as a discrete sum over the azimuthal orders m characterizing the seismic source (see, e.g., *Moinet and Dietrich* [1998] for analytical expressions of displacement components for $m = 0$ and $m = \pm 1$).

[24] The integral over the horizontal wave number is carried out by using the discrete wave number method [Bouchon, 1981], which consists in the introduction of an infinite set of concentric rings of secondary sources distributed at equal radial intervals from the point source $(0, 0, z_s)$. The spatial periodicity L_p of these additional sources, which determines the wave number sampling $\Delta k = 2\pi/L_p$, is chosen such that the disturbances generated by the secondary sources reach the farthest observation point $x_m(r_m, \theta_m, z_m)$ outside the time window of interest $t_{\max}$. This requirement is expressed by the following conditions, which are valid in a homogeneous medium:

$$r_m < L_p/2 \quad \sqrt{(L_p - r_m)^2 + (z_m - z_s)^2} > v_{\max} t_{\max}, \quad (20)$$

where $v_{\max}$ is the largest wave velocity in the medium. Consequently, when considering the propagation of coupled seismic and EM waves, the value of $v_{\max}$ should normally be the largest EM wave velocity, which implies a very large periodicity L_p , a very fine wave number sampling Δk , and therefore unacceptably long computation times. We have numerically verified that when seismo-EM waves are generated at an interface, the computed synthetic signals are identical whether $v_{\max}$ is based on the largest seismic or EM wave velocity. This observation results from the characteristics of the radiation patterns of seismo-EM converted waves, which show a rapid falloff of the amplitudes as a function of source-receiver offset. Consequently, the amplitudes of EM waves generated at an interface by secondary sources, measured along the receiver array, are always negligibly small compared to the EM wave radiation induced by the principal source.

[25] The integral over frequency is computed by using a discrete Fourier transform with complex frequencies, a procedure which is described in detail by Bouchon [1979]. Finally, the medium response derived in cylindrical coordinates (r, θ, z) is expressed in Cartesian coordinates (x, y, z) by using classical rotation formulas for each wave field.

3. Reciprocity Tests

[26] The numerical checking of the reciprocity of Green's tensors is a good way to track possible errors in our simulation program such as missing or additional events, incorrect amplitudes or waveforms, and polarity errors. Let $\mathbf{G}_j^i(\mathbf{x}_r, t; \mathbf{x}_s, t_0)$ ($i = 1, 3; j = 1, 3$) be the Green's tensor representing seismic or EM wave fields along direction i , recorded at location $\mathbf{x}_r$ and measured at time t which are generated by a point source oriented in direction j , located at

position $\mathbf{x}_s$ and acting at time t_0 . The reciprocity theorem of Green's tensors stipulates that

$$\mathbf{G}_j^i(\mathbf{x}_r, t; \mathbf{x}_s, t_0) = \mathbf{G}_i^j(\mathbf{x}_s, t; \mathbf{x}_r, t_0). \quad (21)$$

We will show that our simulation program verifies this relationship for both seismic and EM waves when they are considered independently, but also for the coupled seismo-EM waves, for which *Pride and Haartsen* [1996] analytically demonstrated the reciprocity relations.

3.1. Seismic Displacements

[27] In order to verify the reciprocity relations (20) for the seismic displacements generated by mechanical point sources, we consider a simple model consisting of a 100-m-thick porous layer (l_1) overlying a porous half-space (l_2). The properties of the l_1 and l_2 materials are displayed in Table 1. The two media mainly differ by their porosity, permeability, and mechanical properties of the rock frame. The complex velocities of the *PSVTM* waves in the two layers are displayed in Table 2 at a frequency of 70 Hz. The real part of the complex velocity v corresponds to the real wave velocity, whereas the imaginary part is linked to the wave attenuation $\alpha = \omega/\Im(v)$.

[28] The reciprocity was checked by considering three point forces F_x, F_y , and F_z oriented along the x, y , and z axes, respectively, of our Cartesian coordinates system and two reciprocal source-receivers configurations, as shown in Figure 1. In the first configuration (Figure 1a) the seismic source is located near the free surface ($z_s = 0.1$ m). A three-component seismic receiver array is located in the lower half-space at depth $z_r = 150$ m and lateral offset $y = 200$ m in order to record the full *PSVSH* motions. This array consists of 30 receivers regularly spaced between 0 and 500 m along the x axis. The second configuration (Figure 1b) is the reciprocal of the previous one, i.e., the source is located at depth $z_s = 150$ m with lateral offset $y = -200$ m and the receivers are placed at depth $z_r = 0.1$ m at distances ranging between $x = -500$ m and $x = 0$ m.

[29] All seismograms were computed over a duration of 0.6 s. The source time function used is a zero-phase Ricker wavelet having a dominant frequency of ~ 70 Hz. The magnitude of each point force was set to $F = 7.85 \times 10^3$ N, which approximately corresponds to a stack of 100 blows of a 4-kg hammer accelerating at 2 g ($g \simeq 9.81$ m/s²).

[30] The seismograms displayed in Figure 2a show the inline displacements u_x generated by a vertical point force F_z with the "down" source-receiver geometry (receivers below the source, Figure 1a). The value displayed in the middle of the section is the global maximum amplitude measured in the section. Figure 2a shows the *P* and *S* direct waves between the source and the receivers, small-amplitude *PS* and *SP* converted waves at the interface at 100 m depth, a *P*-to-*S* converted wave at the free surface (*PFS*), and multiple *S* reflections between the interface and the free surface (*SS* and *SSS*). The same events can also be identified in Figures 2b–2d, although with different amplitudes. Figure 2b presents the corresponding "up" source-receiver geometry (receivers above the source). A comparison of Figures 2a and 2b shows that the inline horizontal displacements u_x generated by a vertical point force F_z are exactly identical to the vertical displacements u_z generated by an

Table 1. Properties of the Homogeneous Porous Layers Used in the Numerical Calculations

Properties	l_1	l_2	Sand	Sandstone
Porosity ϕ , %	15	30	30	20
dc permeability k_0 , m ²	10^{-12}	10^{-11}	10^{-11}	10^{-12}
Solid bulk modulus k_s , GPa	36	36	35	36
Fluid bulk modulus k_f , GPa	2.2	2.2	2.2	2.2
Frame bulk modulus k_{fs} , GPa	9.0	7.0	0.4	5.0
Frame shear modulus G_{fs} , GPa	7.0	5.0	0.5	7.0
Fluid viscosity η , Pa s	10^{-3}	10^{-3}	10^{-3}	10^{-3}
Solid density ρ_s , kg/m ³	2700	2700	2600	2700
Fluid density ρ_f , kg/m ³	10^3	10^3	10^3	10^3
Salinity C_0 , mol/L	10^{-3}	10^{-3}	10^{-3}	10^{-3}
Temperature T , K	298	298	298	298
Fluid permittivity ϵ_f	80	80	80	80
Solid permittivity ϵ_s	4	4	4	4
Tortuosity α_∞	3	3	3	3

inline horizontal point force F_x , in terms of waveform, signal polarity and amplitude. Similarly, Figures 2c and 2d show that the vertical displacements u_z generated by a crossline horizontal point force F_y are equivalent to the crossline horizontal displacements u_y generated by a vertical point force F_z . All combinations of displacements and forces were successfully tested in this way, leading to the conclusion that our program verifies the reciprocity relationships for seismic excitations and seismic receivers within a poroelastic stratified medium.

3.2. Electromagnetic Wave Fields

[31] Electromagnetic wave fields generated by electric dipoles must also verify the reciprocity relationships in equation (20). To test this case, we consider a three-layer stratified medium, in which a 5-m-thick sand layer is embedded between an upper half-space (air) and a lower half-space representing a clayey material. The constitutive properties influencing the EM wave propagation are displayed in Table 3 for each layer.

[32] As in the seismic case, two reciprocal source-receiver configurations are considered. The first one corresponds to an upward propagation from source to receivers. In this configuration, 30 receivers are regularly spaced between 0 and 10 m along the x axis at a depth of 1 cm below the air-sand interface with an offset of 1 m along the y axis. An emitting electric dipole, oriented either along the x axis or the y axis, is located at 4 m depth inside the sand layer, 1 m above the sand-clay interface. This geometry mixes the TE and TM propagation modes. The current density source is created by a 1-m-long dipole antenna carrying a 1-A current with a 200-MHz center frequency. The reciprocal geometry corresponds, as before, to a downward propagation from source to receivers.

[33] The electrograms E_x presented in Figure 3a exhibit three main events generated by an electric antenna D_y . At most locations the first arrival is a very weak air-refracted

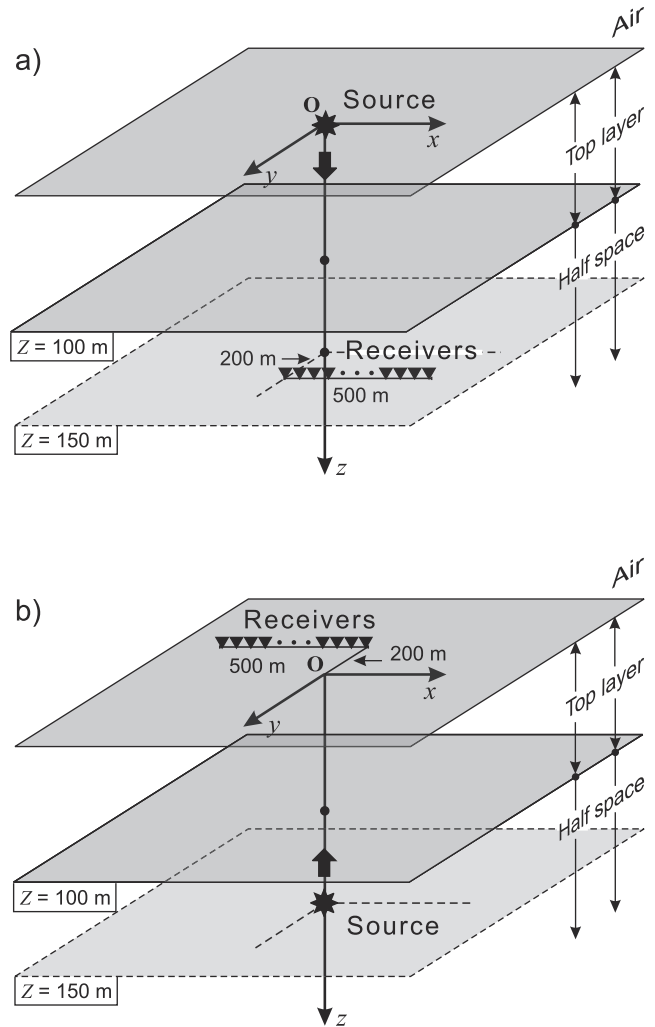


Figure 1. Source-receiver configurations used to test the reciprocity of Green's functions for seismic displacements and point sources. (a) Receivers below the source ("down"). (b) Receivers above the source ("up").

wave. It is followed by the direct wave propagating from the source to the receivers within the sand layer. The latter event corresponds to an EM wave reflected from the interface separating the sand and clay layers. The four sections displayed in Figure 3 show that reciprocity is verified for EM waves generated by EM sources in the case considered.

3.3. Seismo-EM Wave Fields

[34] *Pride* [1994, Appendix B] demonstrates that his theoretical model verifies the Onsager reciprocity, which states that the electrokinetic and electro-osmotic coupling coefficients L_{12} and L_{21} are equal at all frequencies, provided that the electric double layer is thin relative to

Table 2. Complex Velocities ($\Re$, $\Im$) of the $PSVTM$ Wave Types

Velocity, m/s	l_1 (70 Hz)	l_2 (70 Hz)	Sand (100 Hz)	Sandstone (100 Hz)
P_f wave	(3195, -0.0052)	(2865, -0.34)	(1860, -1.89)	(2992, -0.0056)
P_s wave	(44.1, -43.1)	(108.8, -101.6)	(56.6, -51.3)	(44.9, -43.3)
SV wave	(1692, -0.15)	(1511, -1.5)	(486, -0.71)	(1722, -0.23)
TM wave $\times 10^5$	(6.15, -6.07)	(4.36, -4.31)	(5.22, -5.13)	(6.4, -6.25)

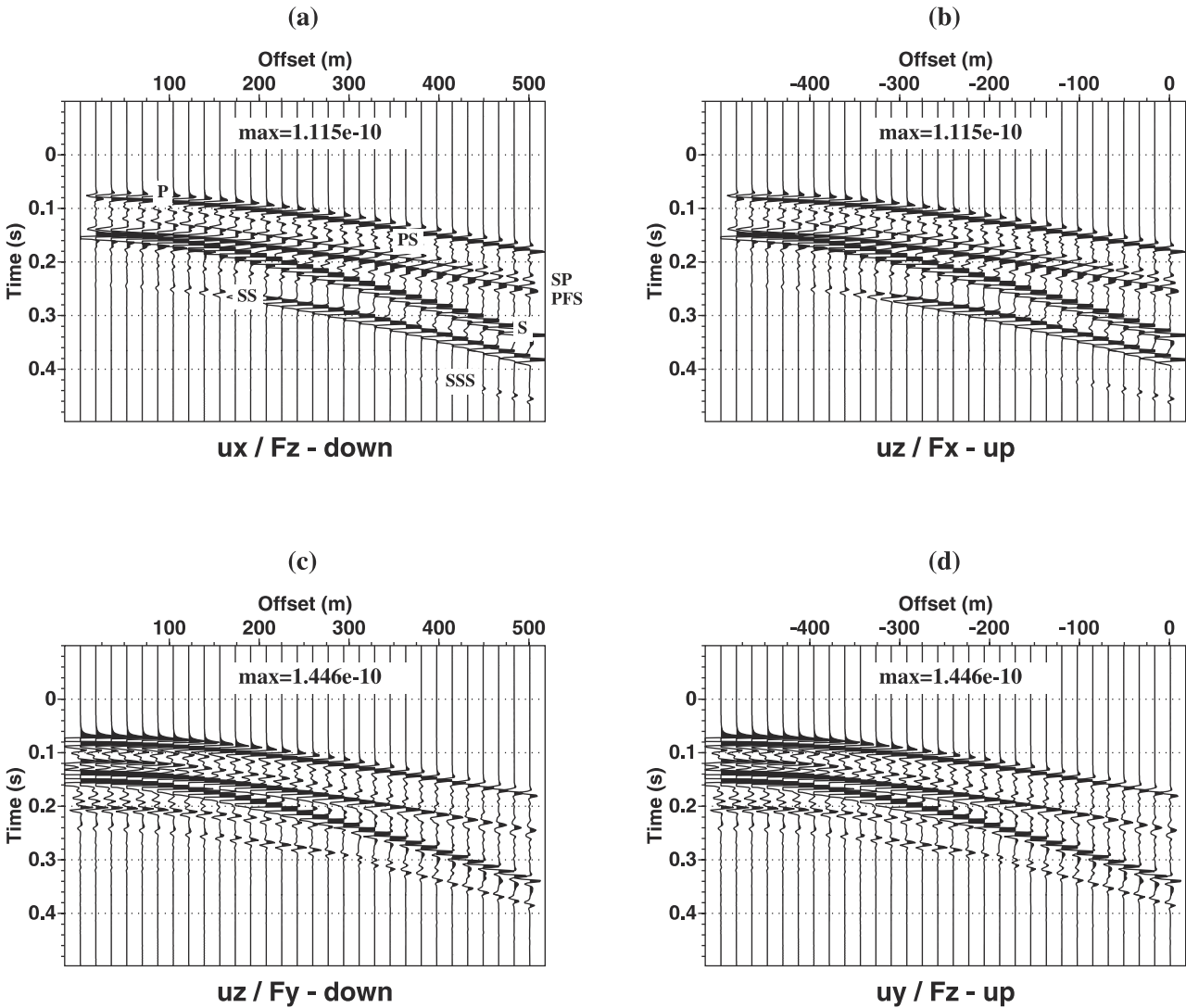


Figure 2. Reciprocity of seismic displacements generated by various seismic point sources. The reciprocity is verified between Figures 2a and 2b and between Figures 2c and 2d. The maximum amplitude (m) is given in each panel.

the grain size. This symmetry property has sometimes been challenged by experimental measurements. *Auriault and Strzelecki* [1981] demonstrated that L_{21} can be a thousand times as high as L_{12} for kaolinite clay samples. In other experiments performed with kaolinite, *Mitchell* [1976] measures L_{12}/L_{21} ratios ranging between 0.88 and 1 and concludes that the coefficients L_{12} and L_{21} are almost equal. Consequently, the model developed by *Pride* [1994] should be considered with care when considering clayey materials.

[35] In this section we study the conversions between seismic and EM waves at an interface located at 100 m depth but do not consider the local electric and magnetic fields accompanying seismic waves or the seismic displacements locally generated by the propagation of EM waves. It has been demonstrated by *Garambois* [1999] that reciprocity is also verified in these cases.

[36] The only difference between the model used in this study and the model presented in Figure 1 concerns the locations of the source and receivers, which are now at 0

Table 3. Material Properties Used in the Computation of Radargrams (EM Wave Propagation)

Medium	ϕ , %	κ_f	κ_g	α_∞	C_0 , mol/L	κ	σ_0 , S/m	V_{EM} at 200 MHz, m/ns
Air	1	1	1	1	0	1	0	(0.3, 0)
Sand	30	80	4	3	10^{-3}	26	9.3×10^{-4}	(0.058, -9.3×10^{-5})
Clay	10	80	4	3	10^{-1}	6.5	3×10^{-2}	(0.11, -0.022)

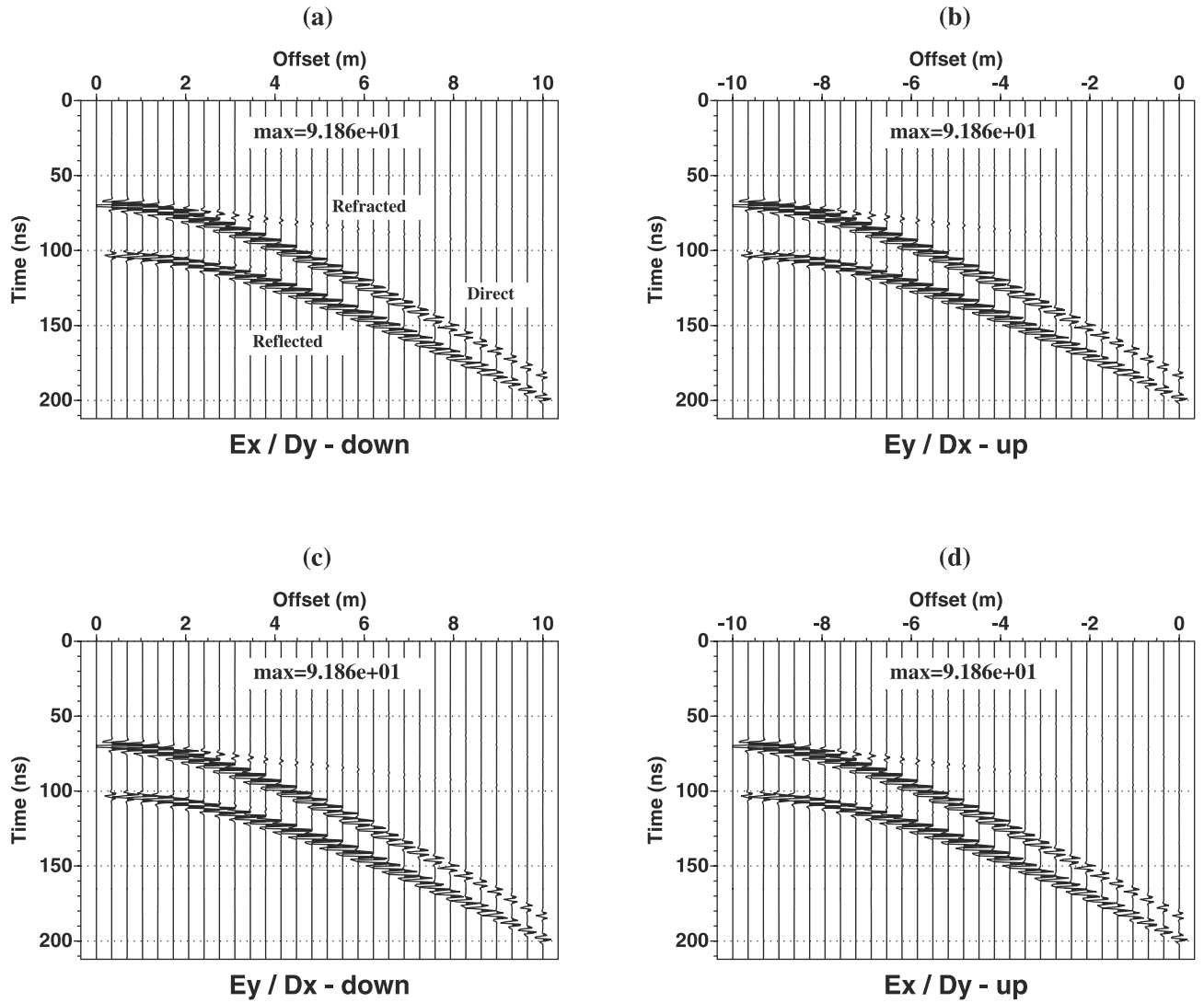


Figure 3. Symmetry of the horizontal electric components of EM waves. The maximum amplitude (V/m) is given in each panel.

and 200 m depth and vice versa in the reciprocal geometry. The receivers are spread out from 0 to 200 m in the inline direction x and offset by 20 m in the crossline direction y . The dominant frequency of the source is 100 Hz.

[37] Figure 4 shows that reciprocity is verified for different combinations of point sources and wave fields. For instance, the perfect symmetry between Figures 4a and 4b directly derives from equation (50) of *Pride and Haartsen* [1996]. Figure 4a displays the total vertical seismic displacements corresponding to seismic P (P -EM) and S (S -EM) waves generated at the interface at 100 m depth by electro-osmotic conversions from an incident EM wave created by a horizontal electric dipole oriented along the x axis. It can be seen in Figure 4b that these displacements (in meters) are exactly equivalent to the electric fields (in V/m) contained in the EM waves generated at the same interface by electro-kinetic conversions of seismic P (EM - P) and S (EM - S) waves created by a vertical point force. Note that in this example, only the transmission coefficients t_U^{SVTM} , t_U^{PTM} ,

t_U^{SHTE} , t_D^{TMSV} , t_D^{TMPf} , and t_D^{TESH} were accounted for, in order to avoid the overwhelming predominance of local seismic or EM fields accompanying the EM or seismic wave propagation.

[38] Figures 4c and 4d display another combination of wave fields and sources, namely, the symmetry between the horizontal displacements u_x generated at the interface by EM waves created by an electric dipole D_y , and the horizontal electric fields E_y generated by seismic waves created by a seismic point force F_x . All source-receiver combinations were successfully tested, thus showing the consistency of the formulation used.

[39] Finally, we present in Figure 5 the individual contributions of the vertical grain displacement u_z (Figure 5a) and of the relative fluid-to-solid vertical displacement w_z (Figure 5b) to the total vertical displacement displayed in Figure 4a, which is the quantity measured by geophones in the field. It appears in this example that the contribution of the relative fluid-to-solid displacement is extremely weak (<1% of the total displacement). This justifies the approx-

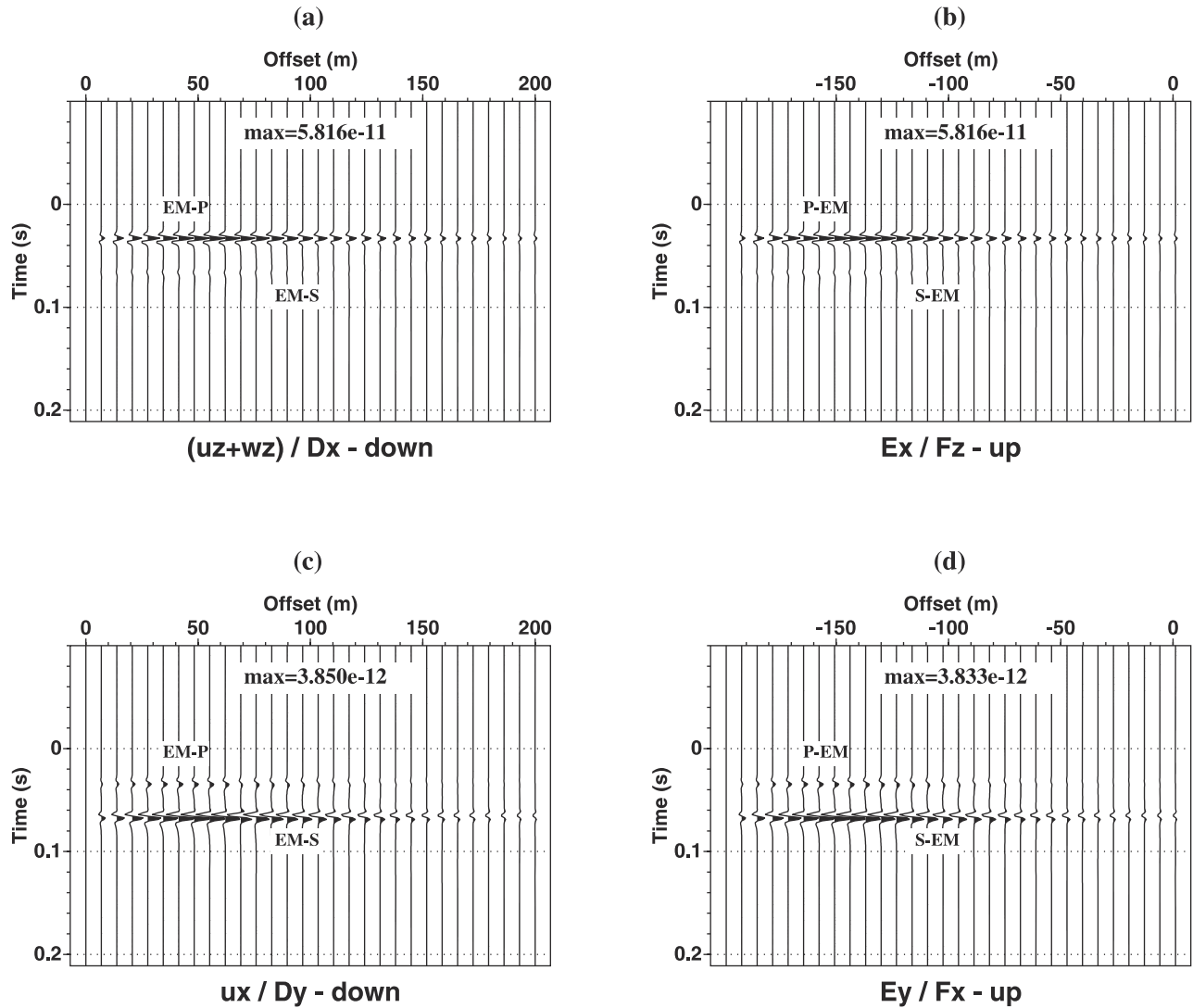


Figure 4. Reciprocity of seismo-electromagnetic wave fields. (a) Total vertical seismic displacement created by electro-osmotic conversion of an EM wave propagating downward through the interface at 100 m depth. (b) Horizontal component of the electric field of the EM waves created by electrokinetic conversion of P and S waves at the same interface. (c) and (d) Horizontal grain displacement u_x and transverse electric field E_y , respectively generated by an electric dipole D_y and by a mechanical force F_x in the two reciprocal configurations considered.

imation made by *Garambois and Dietrich* [2001] concerning the calculation of the theoretical transfer function between seismic and electric or magnetic signals, where the contribution of fluid-to-solid displacements was neglected.

4. Propagation Snapshots

[40] Snapshots of seismic displacements and electric and magnetic fields corresponding to the coupled seismic and EM wave propagation in porous media were computed by using a two-dimensional grid of receivers in the vertical plane of propagation. These snapshots allow us to illustrate some properties of electrokinetic coupling by following the propagation of seismic waves as a function of time in a layered porous medium and by analyzing the EM disturbances that they are generating.

[41] The model used in this study is shown in Figure 6a. It consists of a 100-m-thick consolidated sand layer embedded between two identical layers. The properties and associated wave velocities characterizing the two identical layers and the consolidated sand layer are those of layers l_1 and l_2 presented in Tables 1 and 2, respectively. To compute the snapshots, we used a grid of 199×50 receivers regularly spaced 5 m apart, between -495 m and 495 m in the horizontal direction and between 20 and 265 m in the vertical direction. The seismic source is located at 1 m depth in the middle of the receiver array. The calculations were performed with a vertical point force generating both P and S waves at a dominant frequency of 170 Hz. Free surface reflections were included in the computations.

[42] Figure 6b displays the wave fronts associated with the horizontal displacements generated by the seismic source. Figure 6b (top snapshot) ($t = 32.8$ ms) shows

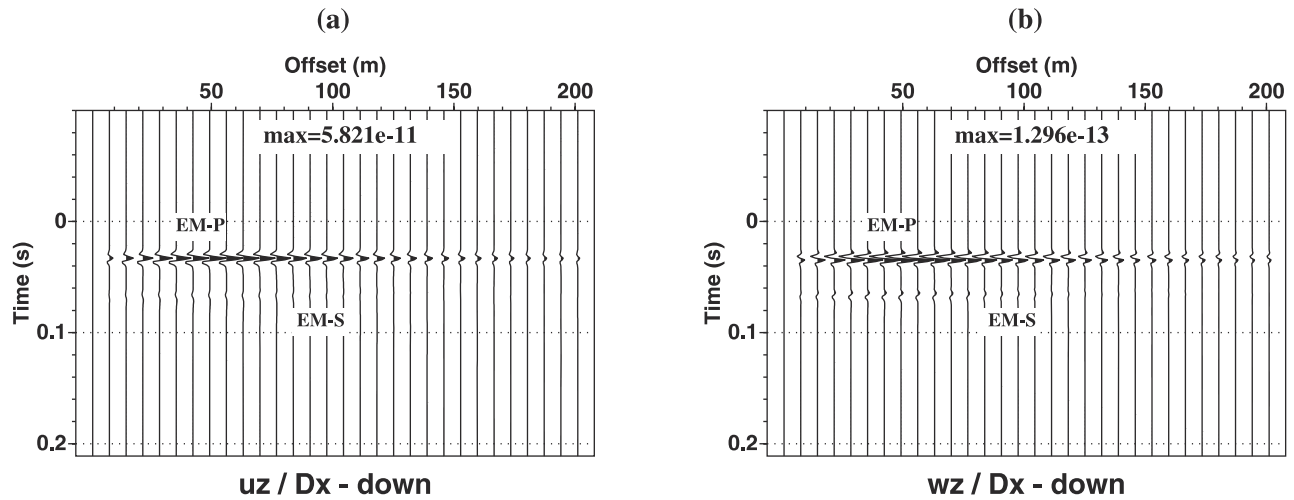


Figure 5. Comparison of the waveforms and amplitudes of (a) the vertical solid grain displacement u_z and (b) the relative fluid to solid displacement w_z . These displacements are generated by electro-osmotic conversions of a downward traveling EM wave into seismic P and S waves at an interface.

the P and S wave fronts in the top layer at the time the P wave reaches the first interface at 103 m depth. The second snapshot ($t = 68$ ms) shows the direct S wave arriving at the first interface as well as the P wave transmitted through the first interface and reaching the second interface at 203 m depth. The second snapshot also exhibits the primary P wave reflected upward off the first interface, as well as two P to S converted waves generated on both sides of the first interface. The bottom two snapshots computed at times $t = 100$ ms and $t = 132$ ms are increasingly complex because of the numerous reflected,

transmitted, and mode-converted waves created at the two interfaces and at the free surface.

[43] We have similarly computed the electric (Figure 7a) and magnetic fields (Figure 7b) generated by the same vertical point force. The resulting snapshots clearly show the electric and magnetic fields locally induced by the propagation of P and S waves through the fluid-saturated porous medium. A comparison with Figure 6b shows that there is a natural decomposition of the EM response according to the seismic wave type. Indeed, one can observe that the electric fields are only associated with

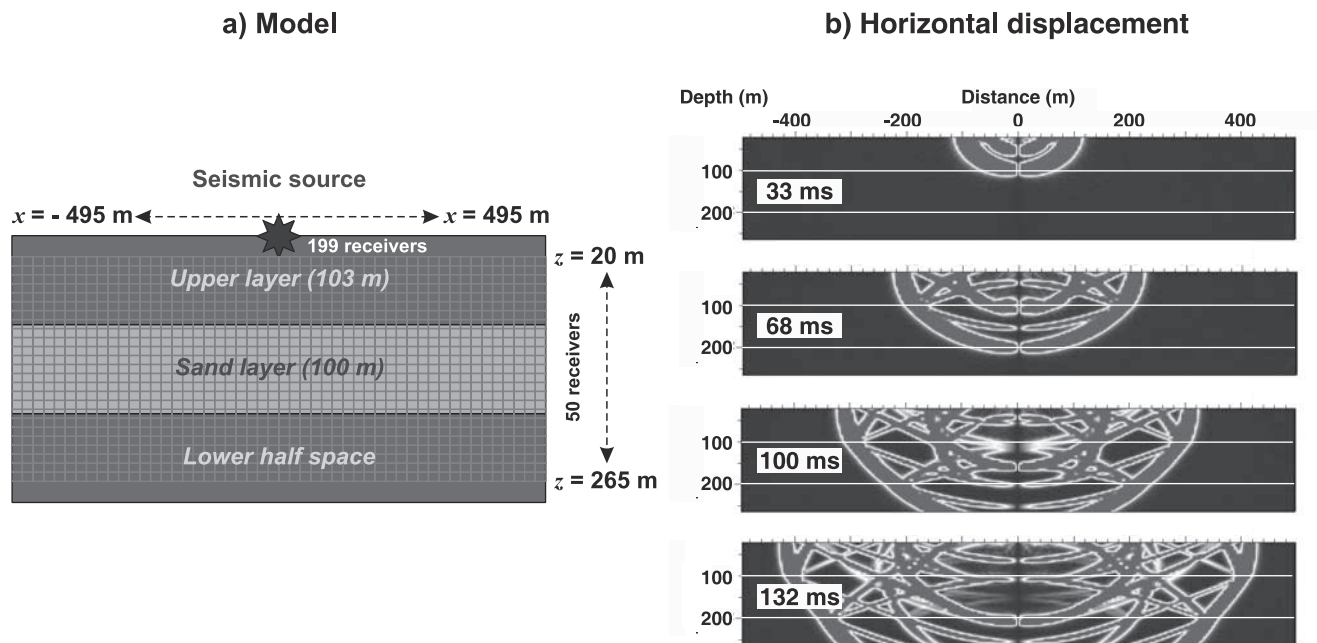


Figure 6. (a) Model used for the computation of snapshots of seismic, electric, and magnetic wave fields. (b) Wave fronts of horizontal displacements generated by a vertical point force at times $t = 32.8$ ms, $t = 68$ ms, $t = 100$ ms, and $t = 132$ ms. See color version of this figure at back of this issue.

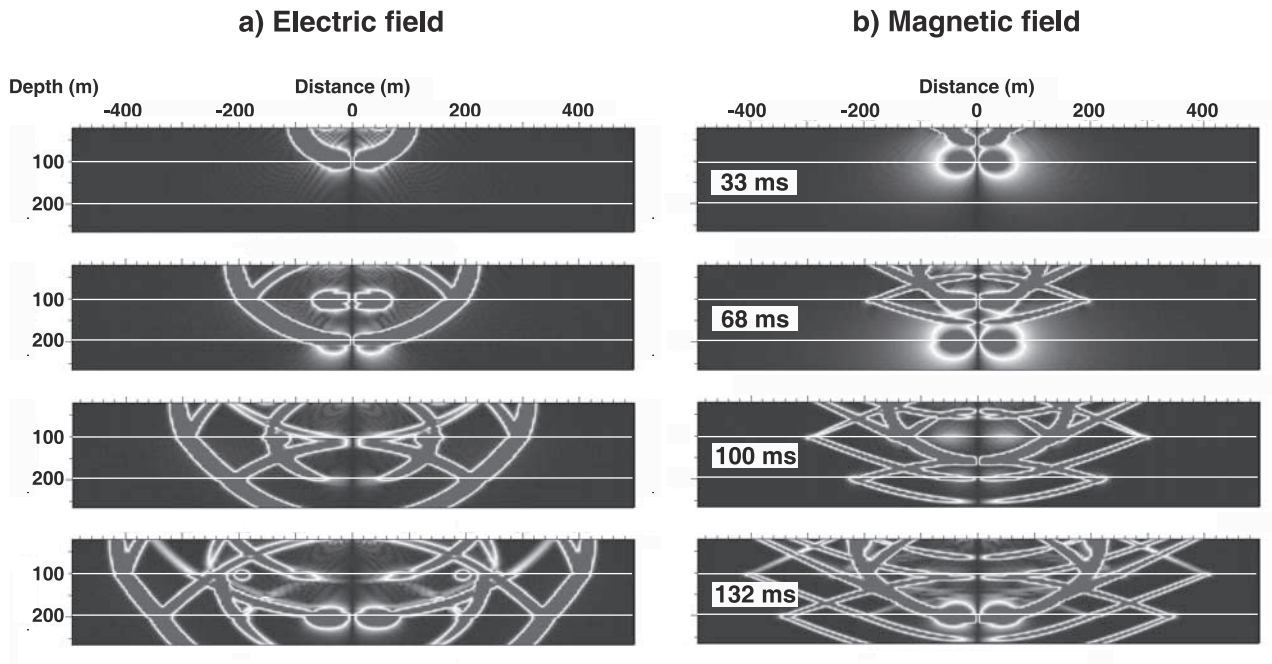


Figure 7. Snapshots of the wave fronts of (a) the electric field E_x and (b) the magnetic field H_y generated by a vertical point force at times $t = 32.8$ ms, $t = 68$ ms, $t = 100$ ms, and $t = 132$ ms. See color version of this figure at back of this issue.

the propagation of P waves, whereas the magnetic fields are uniquely associated with the propagation of S waves, except at the interfaces. It is also seen that the electric field generated by induction of the magnetic field accompanying S waves is negligible. These observations confirm that electric antennas can act as P wave detectors and that magnetometers could be used as S wave sensors. The transfer functions between seismic and electric fields on one hand and seismic and magnetic fields on the other hand, as well as their sensitivity to the medium properties, were extensively studied by *Garambois and Dietrich* [2001].

[44] A second-order effect, which consists of bipolar EM radiations centered around each interface, appears simultaneously in the snapshots of electric and magnetic fields each time a seismic P or S wave crosses the interfaces. This effect was experimentally identified by *Martner and Sparks* [1959] and by *Garambois and Dietrich* [2001] for shallow interfaces. *Thompson and Gist* [1993] also used these seismo-EM converted waves to image the subsurface down to 300 m depth.

[45] Using our simulation code, we will now concentrate on the EM disturbances generated at depth to study some of their properties (radiation pattern, amplitude, sensitivity) and to assess their potential applications for the characterization of the subsurface and its fluid content.

5. Properties of Seismo-EM Wave Conversions at Depth

5.1. Fresnel Zones

[46] The representation theorem indicates that each point of a reflecting surface behaves like a secondary source.

Consequently, a wave reflection, with or without mode conversion, can be computed by integrating the contribution of point scatterers over a certain area. In the elastic case this representation was recently demonstrated by *Moinet and Dietrich* [1998]. The first Fresnel zone is defined as a region of constructive interferences centered around the specular reflection point given by Snell's law. In order to determine the shape and dimensions of the surface which effectively contributes to seismo-EM conversions at an interface, we extend the concept of Fresnel zone to include these specific mode-converted reflections by using formulas originally derived by *Eaton et al.* [1991] for P - SV reflections.

[47] In the general case where source and receiver are not coincident, the form of the first Fresnel zone corresponding to a mode-converted reflection is an oval containing the specular reflection point, elongated in the source-receiver direction and symmetric in respect to that axis. This oval can be described by three parameters defined from the specular reflection point, as described by *Eaton et al.* [1991]: the inline radii ϵ_- and ϵ_+ and the crossline radius ϵ_χ . A straightforward substitution of the S wave velocity by the EM wave velocity in equations (11) and (12) given by *Eaton et al.* [1991] allows us to compute the parameters describing the first Fresnel zone of P -EM (electrokinetic) and EM- P (electro-osmotic) mode-converted reflections. We solved the equations for a 100-m-thick sand layer overlying a sandstone half-space and for a frequency of 100 Hz. The "sand" and "sandstone" properties are those listed in Table 1. The corresponding $PSVTM$ velocities are displayed in Table 2.

[48] Figure 8 presents the first Fresnel zones associated with seismic P to EM (SE), and EM to seismic P (ES)

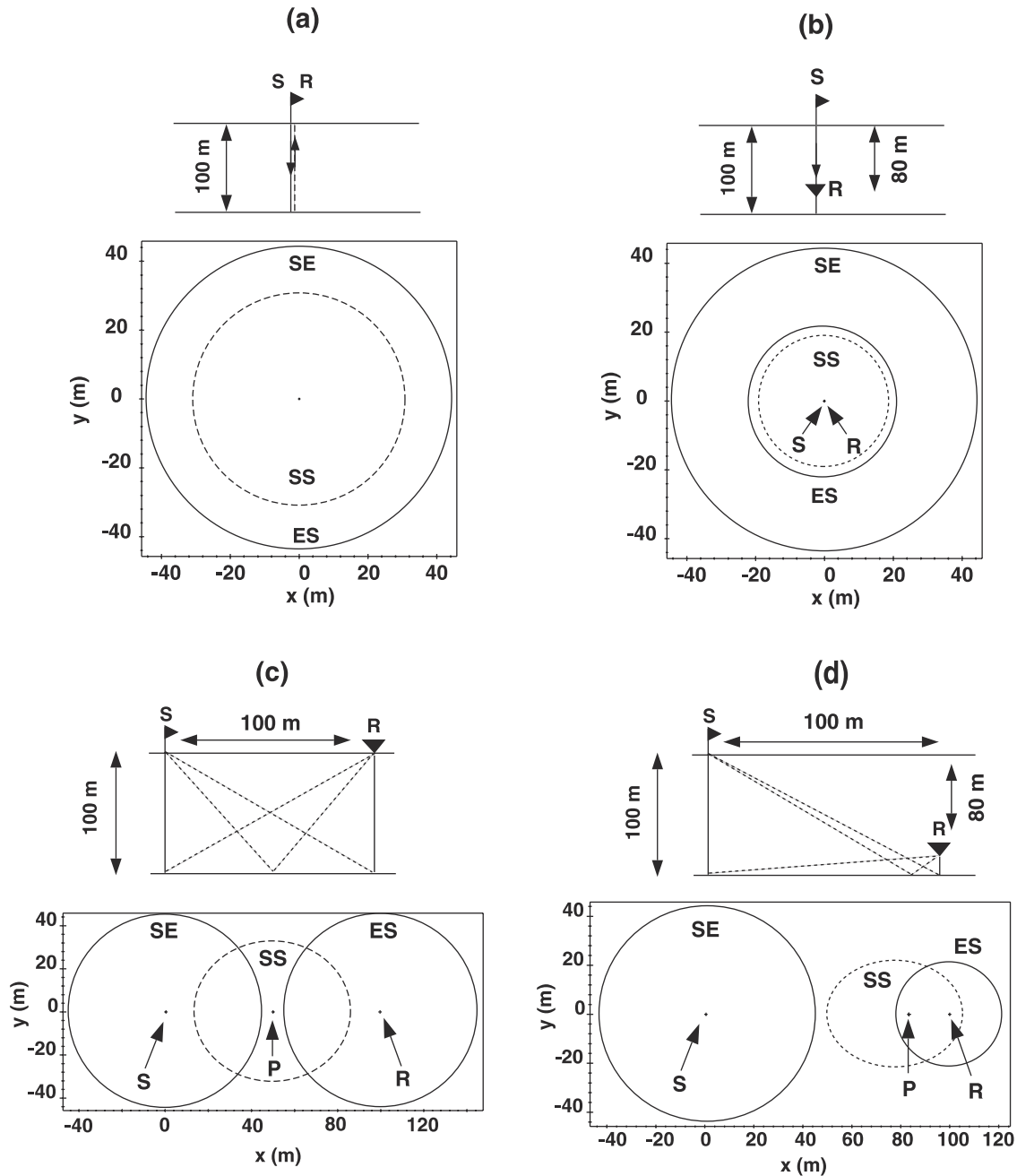


Figure 8. First Fresnel zones associated with seismic P to EM converted waves (labeled SE), with EM to seismic P converted waves (labeled ES) and with seismic P primary reflections (labeled SS). (a) Coincident source S and receiver R at the surface. (b) Buried receiver at 80 m depth and zero offset. (c) Surface configuration with 100 m offset between source and receiver. (d) Borehole configuration with receiver at 80 m depth and horizontal source-receiver offset of 100 m. Point P denotes the specular reflection locus of seismic (SS) reflections.

converted waves, and with the seismic P - P reflection (SS), for different source-receiver configurations. As expected, when source and receiver are coincident, the shape of the first Fresnel zone is circular. The major observations that can be drawn from Figure 8 are that (1) the Fresnel zone associated with seismic to EM conversions (SE) is always centered almost directly under the source, whatever the source-receiver geometry considered; (2) the Fresnel zone associated with EM to seismic conversions (ES) is always

centered almost directly under the receiver; and (3) these two Fresnel zones are always greater in surface than the Fresnel zone associated with a conventional seismic reflection (this is particularly true for SE conversions). Thus, in terms of imaging, the spatial resolution of seismo-EM converted waves at a given depth is smaller than the resolution of pure seismic reflections at the same depth. Conversely, the size of the SE Fresnel zone emphasizes the fact that the very small amplitude coupling phenomena

taking place at the pore level are integrated over a relatively larger area, thus facilitating the observation of EM disturbances at the macroscopic scale, provided that the interface be sufficiently smooth over the first Fresnel zone.

[49] It is useful to note that the results presented here hold true as long as the distance between the seismic source and the discontinuity generating seismo-EM conversions remains greater than the incident wavelength. When this assumption breaks down, the radiation patterns of the seismo-EM waves generated at depth may be more complex than the simple dipole representation discussed below. This case will be considered in further studies.

5.2. Radiation Patterns

[50] As pointed out by *Thompson and Gist* [1993], the seismo-EM response of a plane interface separating two porous layers can be approximated by the radiation of a vertical dipole centered under the seismic source. The purpose of this section is to verify this property. We will also estimate the maximum amplitudes of the electric and magnetic components of the EM waves generated at depth.

[51] For this, we analyze the EM disturbances generated at a sand-sandstone interface by a shot of 1 kg dynamite near the surface. We use the expression of *Haartsen and Pride* [1997] to compute the moment M_0 released by the explosion:

$$M_0 = \sqrt{\frac{4\pi}{3}} \beta^3 K_G J_c \alpha^{3p+1} \epsilon, \quad (22)$$

where α is the mass of explosives, K_G is the Gassmann modulus (a deformation parameter of the porous medium), J_c is the amount of chemical energy released in the explosion, and ϵ is the fraction of J_c converted into elastic waves. The parameters β and p are used to express the source radius as a function of α . In our numerical simulations the values $\alpha = 1$ kg, $p = 1$, $\beta = 1$ m/kg, $K_G = 6.83$ GPa, $J_c = 4.5$ MJ/kg, and $\epsilon = 0.5\%$ were adopted, leading to $M_0 = 2.54 \times 10^7$ J.

[52] Figures 9a, 9b, 9c, and 9d display the vertical and horizontal displacements, the electric field E_x , and magnetic field H_y generated by a near-surface explosion whose dominant frequency is around 100 Hz, respectively. The wave fields were computed for 61 receivers located along the surface every 5 m from -150 to 150 m, symmetrically with respect to the source. A sand-sandstone interface is located at 100 m depth. Direct waves and free surface reflections were omitted in this calculation for sake of clarity. The reflection coefficients corresponding to P -to-EM wave conversions were amplified by a factor of 500 to improve the visibility of these arrivals in the time sections. Figures 9a–9d display the P - P and P - SV reflections at the interface, the electric and magnetic fields that propagate within these two waves, and the EM wave generated by the incident P wave when it propagated through the interface (P -EM). In this case, the amplitude of the electric field propagating within the P - P reflected wave reaches 150 μ V/m and the amplitude of the magnetic field propagating within the P - SV reflected wave reaches 1 μ A/m.

[53] The horizontal component E_x of the electric field radiated by a vertical electric dipole located at the origin of a Cartesian coordinate system is expressed by

$$E_x = \frac{d}{4\pi\epsilon} \frac{3zx}{r^5}, \quad (23)$$

where $r = \sqrt{x^2 + z^2}$. Parameter d denotes the bipolar moment of the dipole; ϵ is the dielectric permittivity of the medium. Figure 10 displays the amplitude versus offset behavior of the horizontal (inline) electric fields measured at the surface associated with the EM disturbance generated by an incident P wave at the sand-sandstone interface (solid lines) and with a vertical electric dipole located under the seismic source at the same depth as the interface. The amplitudes curves are presented for two interfaces at 80 m and 100 m depth. For the sake of the comparison, the moment of the electric dipole was adjusted to $d = 9.65 \times 10^{-11}$ C m. Overall, the two radiation patterns are in good agreement, thus confirming that the contribution of the first Fresnel zone can be reasonably approximated by a vertical electric dipole located under the seismic source. The curves also show that the decrease of the electric field amplitude as a function of source-receiver offset is slightly faster for the complete numerical solution than it is for the simple dipole approximation. This fact may result from the influence of electric multipoles describing the higher-order Fresnel zones, whose influence increases as the distance separating the seismic source and the interface becomes comparable to or lower than the dimension of the first Fresnel zone.

[54] The amplitude level of the seismo-EM disturbances predicted by our numerical simulations (up to 0.15 μ V/m with the parameters used in this study) should normally be detectable in the field with the currently available recording instruments after stacking a number of records and filtering the dominant effect, i.e., the electric and magnetic field propagating within seismic waves. Note that the signal-to-noise ratio can be enhanced by increasing the source energy which, according to equation (22), varies as the square of the mass of dynamite fired. The signal-to-noise ratio will also improve if the conversion interfaces are located at shallow depths. Thus, for an interface at 50 m depth the peak amplitude of the EM signals amounts to $E_x = 0.9$ μ V/m and $H_y = 1.51 \times 10^{-8}$ A/m. These values drop to $E_x = 0.018$ μ V/m and $H_y = 1.06 \times 10^{-9}$ A/m if the interface is located at 150 m depth. Finally, it should be noted that the comparison between the theoretical radiation pattern corresponding to a single dipole and the radiation pattern of EM disturbances recorded in the field can be used as a fast and simple mean to estimate the depth where these EM disturbances are generated. Moreover, as first suggested by *Thompson and Gist* [1993], asymmetric radiation patterns are indicative of a dipping interface and can be used to assess the dip angle of the conversion interface.

5.3. Sensitivity Study

[55] Since seismo-EM disturbances generated at depth can actually be observed in the field [*Martner and Sparks*, 1959; *Thompson and Gist*, 1993; *Garambois and Dietrich*, 2001], it is especially useful to determine

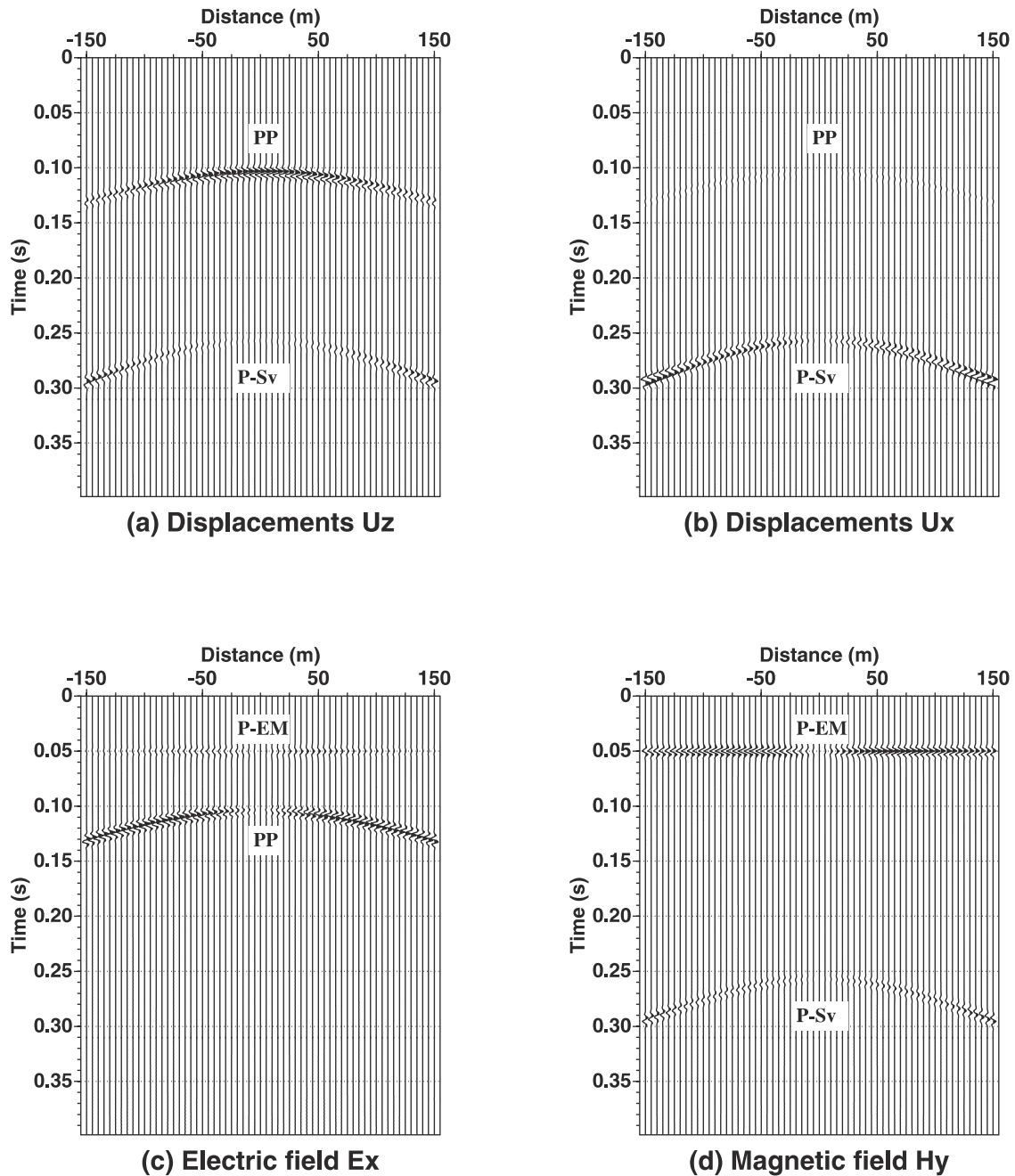


Figure 9. Seismic displacements and electric and magnetic fields generated by a near-surface explosion located in the middle of the profile. The propagation model consists of a 100-m-thick sand layer overlying a sandstone half-space. Direct waves and reflections at the free surface were omitted in the solution. In order to make them more apparent, *P*-to-EM wave conversions were amplified by a factor of 500 in the electric field and magnetic field sections in Figures 9c and 9d.

which properties they are sensitive to. Indeed, the knowledge of the most relevant parameters that determine the magnitude of the seismo-EM waves has direct implications for the characterization of the subsurface and its fluid content by means of an electrokinetic prospecting method such as the one proposed by *Thompson and Gist* [1993]. More generally, this sensitivity study can also be considered as the very first step toward the formulation of the seismo-EM inverse problem which remains to be developed. In their study on electroseismic waves, *Haart-*

sen and Pride [1997] indicated that EM disturbances are generated at contrasts in mechanical or electric (salinity) properties within a fluid-saturated porous medium. In this section we use our full waveform modeling tool to perform a numerical sensitivity study of the EM waves generated at depth.

[56] The sensitivity study is carried out by investigating the amplitude of the EM wave generated at an interface between two porous layers at 50 m depth by an incident *P* wave produced by a 1-kg dynamite shot fired at 10 m depth.

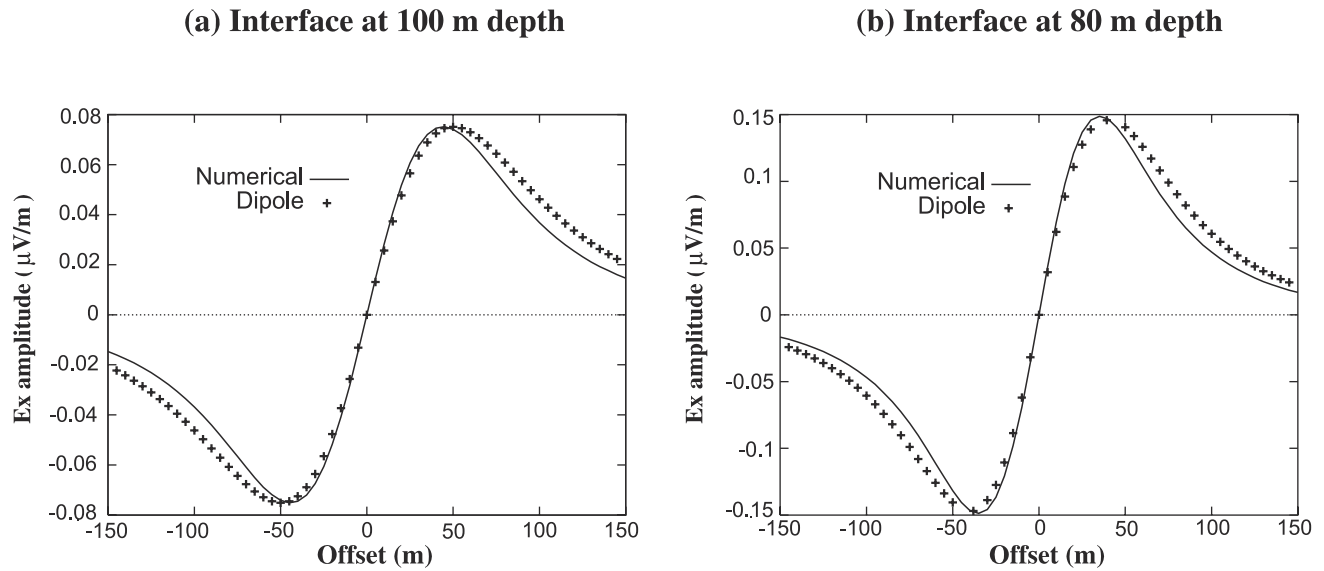


Figure 10. Amplitude versus offset distribution of the horizontal electric field corresponding to an EM disturbance generated by a P wave crossing a sand-sandstone interface (solid lines) and to a vertical electric dipole located directly under the seismic source (crosses). (a) Interface and dipole located at 100 m depth. (b) Interface and dipole located at 80 m depth.

The synthetic seismo-EM waveforms are computed for an array of 26 receivers regularly spaced along the surface between 0 and 250 m from the seismic source. The media above and below the interface are sand layers saturated by salt water whose properties are given in Table 1, except the porosity, which was set to 0.35. The upper layer has fixed properties in order to keep the propagation and attenuation characteristics constant between the surface and the interface. The sensitivity study is carried out by modifying the properties of the lower layer one by one to study their effect on the seismo-EM wave amplitude.

[57] Figure 11 displays the maximum root-mean-square amplitude measured on all traces of the electric E_x and magnetic H_y time sections when the porosity, permeability, fluid salinity, and fluid viscosity are modified in the lower layer. A global sensitivity study performed with respect to all mechanical and electric properties characterizing the lower porous layer indicates that the four above mentioned properties are the most influential parameters in the generation of seismo-EM waves compared to other parameters like shear and bulk frame moduli, density, fluid and solid dielectric permittivity, temperature, and tortuosity. An inspection of the computed amplitudes shows that the EM disturbances are particularly sensitive to strong permeability and salinity contrasts. As expected, the EM amplitude generally increases when the contrast of properties between the two layers increases.

[58] The particular behavior of the EM wave amplitude for small values of porosity in the lower layer can be explained as follows. When the porosity decreases, the charge density of the porous medium decreases as well. Assuming that the interface acts as an electromagnetic dipole, the decrease in porosity results in a strong decrease of the dipole moment in spite of the increase of the porosity contrast between the two layers. A similar behavior is observed for the fluid salinity variations. As first suggested by *Pride and Haartsen* [1996],

when salinity decreases to values where surface conduction in the diffuse layer becomes important, the magnitude of the charge separation is decreased and so is the EM wave amplitude. We can also note that the EM wave generation is favored when the fluid viscosity is low because the fluid circulation and ion mobility are then enhanced, which is essential to EM wave generation.

[59] We do not present the sensitivity results obtained by modifying the other parameters, which are of less importance. As already underlined by *Haartsen and Pride* [1997], contrasts in mechanical properties can also create EM disturbances, especially contrasts in fluid or solid density and contrasts in the shear modulus of the rock matrix. These properties were confirmed by our calculations.

[60] We conclude this sensitivity study with a few remarks. First, it should be noted that the seismo-EM wave amplitude is enhanced when an interface between two porous layers presents contrasts of several properties (and not only one), which is often the case in nature. Second, the sensitivity of the EM waves to permeability and porosity contrasts indicates that seismo-EM waves could be used to detect, characterize, or monitor moderately deep aquifers, as well as fluid-filled fractures and faults at depth. These converted waves could also be used to detect gas-water contacts or other fluid transitions thanks to their strong sensitivity to viscosity contrasts. Finally, the sensitivity of seismo-EM waves to ion concentration could be useful to track pollutants at depth.

6. Conclusion

[61] We have developed and implemented a formulation for the full-waveform modeling of the coupled seismic and electromagnetic wave propagation in fluid-saturated stratified porous media. Our computation scheme is based on the generalized reflection and transmission matrix method of

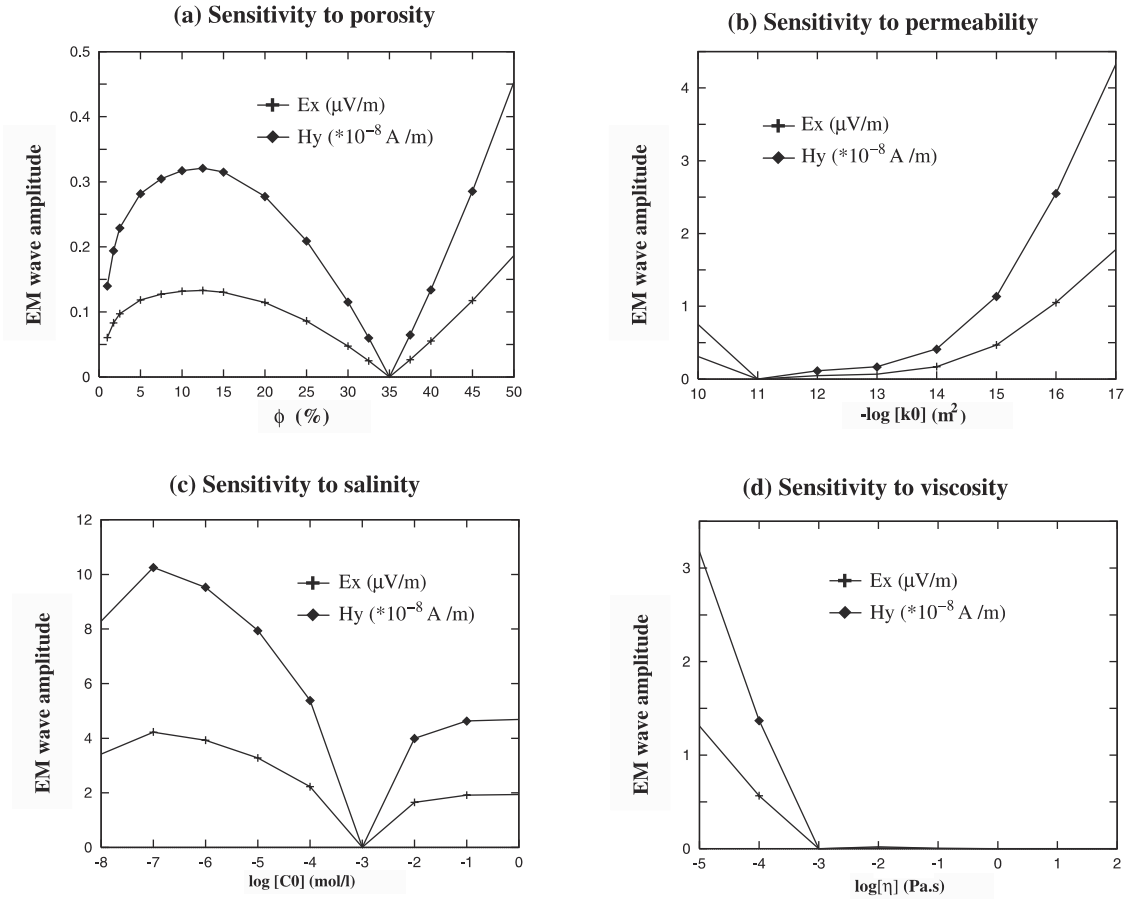


Figure 11. Root-mean-square electric and magnetic field amplitudes of the EM disturbances created at (a) porosity contrasts, (b) permeability contrasts, (c) salinity contrasts, and (d) viscosity contrasts. The contrasts are determined with respect to the properties of the upper layer which are kept constant: $\phi = 35\%$, $k_0 = 10^{-11} \text{ m}^2$, $C_0 = 10^{-3} \text{ mol/L}$, and $\eta = 10^{-3} \text{ Pa s}$.

Kennett and Kerry [1979] and uses the discrete wave number integration method of Bouchon [1981]. We have checked our simulation code by performing reciprocity tests to verify the general reciprocity theorem. In particular, this numerical code can be used to study the seismic and EM wave propagation independently, as a function of the constitutive properties of a fluid-saturated porous medium. Its main application, however, is to simulate the coupled seismic and EM wave propagation and, notably, the EM waves generated by the passage of seismic waves through an interface separating two different media. We have numerically determined the portion of the interfaces which is involved in the generation of the seismo-EM wave conversions, their amplitudes and radiation patterns, and their sensitivity to various contrasts in electric or mechanical properties.

[62] Our studies show that the seismo-EM waves should be detectable with recorders having a high dynamic range, that their amplitude versus offset behavior can be used to estimate their depth of generation, and that they can probably be used to image contrasts of porosity, permeability, fluid salinity, or fluid viscosity at depth. Therefore these waves should prove useful in hydrocarbon exploration, for environmental purposes (aquifer delineation, pollutant tracking), and for crack detection, since all these

applications can theoretically be performed with the resolution of seismic methods.

Appendix A: Representations of Seismic and EM Point Sources

[63] Here we give the expressions of the source discontinuity vector introduced in equation (18) for various mechanical and electric point sources. The most general representation of a point source is a combination of a source moment tensor $\mathbf{M}$ with a simple force $\mathbf{F}$ [see, e.g., Kennett and Kerry, 1979]. In a Cartesian coordinate system (x, y, z) ,

$$f_i = -\partial_j [M_{ij} \delta(\mathbf{x})] + F_i \delta(\mathbf{x}); \quad i, j = x, y, z. \quad (\text{A1})$$

In a cylindrical coordinate system (r, θ, z) , equation (A1) can be rewritten as

$$\begin{bmatrix} f_r \\ f_\theta \\ f_z \end{bmatrix} = \left\{ \begin{pmatrix} M_{rr} & M_{r\theta} & M_{rz} \\ M_{\theta r} & M_{\theta\theta} & M_{\theta z} \\ M_{zr} & M_{z\theta} & M_{zz} \end{pmatrix} \begin{bmatrix} \frac{\partial}{\partial r} \\ \frac{1}{r} \frac{\partial}{\partial \theta} \\ \frac{\partial}{\partial z} \end{bmatrix} + \begin{bmatrix} F_r \\ F_\theta \\ F_z \end{bmatrix} \right\} \frac{\delta(r)}{r} \delta(\theta) \delta(z - z_s). \quad (\text{A2})$$

We assume that each type of source equally acts on the solid matrix and on the fluid.

A1. Explosion

[64] The source terms corresponding to an explosive point source were derived by *Haartsen and Pride* [1997]. Here we give an alternative derivation of the source discontinuity vector. In a homogeneous and isotropic material an explosion can be modeled as an instantaneous relaxation of the elastic deformation energy stored in a spherical volume. An explosion is solely represented by an isotropic tensor, i.e., $M_{ij} = \Phi(t)M_0\delta_{ij}$, where δ_{ij} is the Kronecker symbol, M_0 is the deformation energy introduced in equation (22), and $\Phi(t)$ is the time signature of the seismic energy relaxation, which is often represented by a normalized Ricker pulse.

[65] Hence, in a cylindrical coordinate system (r, θ, z) the forces generated by an explosion at depth $z = z_s$ are written as

$$\begin{aligned} f_r &= M_0\Phi(t)\left\{\frac{\partial}{\partial r}\left[\frac{\delta(r)}{r}\right]\right\}\delta(\theta)\delta(z-z_s), \\ f_\theta &= M_0\Phi(t)\frac{\delta(r)}{r^2}\left\{\frac{\partial}{\partial\theta}[\delta(\theta)]\right\}\delta(z-z_s), \\ f_z &= M_0\Phi(t)\frac{\delta(r)}{r}\delta(\theta)\left\{\frac{\partial}{\partial z}[\delta(z-z_s)]\right\}. \end{aligned} \quad (\text{A3})$$

These expressions are transformed into the frequency ω and wave number k domain by using a Fourier-Bessel transform and a normalization of the horizontal components by a factor ik [Kennett, 1983]:

$$\begin{aligned} \hat{f}_1 &= ikM_0\hat{\Phi}(\omega)\delta(z-z_s), \\ \hat{f}_2 &= 0, \\ \hat{f}_z &= M_0\hat{\Phi}(\omega)\frac{\partial}{\partial z}[\delta(z-z_s)], \end{aligned} \quad (\text{A4})$$

where $\hat{\Phi}(\omega)$ is the Fourier transform of $\Phi(t)$.

[66] The general expressions of the force vectors $\mathbf{F}$ introduced in equation (1) for the *PSVTM* and *SHTE* cases are given by *Haartsen and Pride* [1997, equation (37)]. For an explosion they reduce to

$$\mathbf{F}^{PSVTM} = \left[0, 0, \frac{ik}{\omega^2\bar{\rho}}\hat{f}_1, -\hat{f}_1 + \frac{\rho_f}{\bar{\rho}}\hat{f}_1, -\hat{f}_z, -\hat{f}_z, -L\hat{f}_1, 0\right]^T \quad (\text{A5})$$

$$\mathbf{F}^{SHTE} = [0, 0, 0, 0]^T. \quad (\text{A6})$$

The definitions of the parameters $\bar{\rho}$, ρ_f and L are given by *Pride and Haartsen* [1996].

[67] *Kennett* [1983] has shown that a force system of azimuthal order m enclosed in a horizontal plane can be split up into contributions of the form

$$\mathbf{F}(z_s) = \mathbf{F}_1(k, m, z_s, \omega)\delta(z-z_s) + \mathbf{F}_2(k, m, z_s, \omega)\frac{\partial}{\partial z}[\delta(z-z_s)], \quad (\text{A7})$$

thus allowing the source discontinuity vector to be expressed as

$$\mathbf{S}(z_s) = \mathbf{F}_1(k, m, z_s, \omega) + \mathbf{A}(z_s)\mathbf{F}_2(k, m, z_s, \omega), \quad (\text{A8})$$

where $\mathbf{A}(z_s)$, the matrix introduced in equation (1), is detailed by *Haartsen and Pride* [1997]. Using equation (A4) in equation (A5), we obtain the two contributions defined in equation (A7), namely,

$$\begin{aligned} \mathbf{F}_1 &= M_0\hat{\Phi}(\omega)\left[0, 0, -\frac{k^2}{\omega^2\bar{\rho}}, i\left(\frac{\rho_f}{\bar{\rho}} - 1\right)k, 0, 0, -ikL, 0\right] \\ \mathbf{F}_2 &= M_0\hat{\Phi}(\omega)[0, 0, 0, 0, -1, -1, 0, 0]. \end{aligned} \quad (\text{A9})$$

Finally, by developing equation (A8), we get

$$\begin{aligned} \mathbf{S}(z_s)^{PSVTM} &= M_0\hat{\Phi}(\omega)\left[0, \frac{C-M}{\Delta}, \frac{C-H}{\Delta}, 2ikG_{fr}\frac{C-M}{\Delta}, 0, 0, 0, 0\right] \\ \mathbf{S}(z_s)^{SHTE} &= [0, 0, 0, 0]. \end{aligned} \quad (\text{A10})$$

The deformation parameters C , M , H , and Δ are defined by *Pride and Haartsen* [1996]. The parameter G_{fr} is the frame shear modulus of the porous material. Expression (A10) slightly differs from the one presented by *Haartsen and Pride* [1997, equation (C13)] and corrects a minor error in their article.

A2. Simple Oriented Forces

[68] We consider point forces oriented in the x , y , and z positive directions, whose amplitudes are F_x , F_y , and F_z , respectively. Using the same method as above, we obtain the following expressions of the source vector discontinuities for azimuthal orders $m = \pm 1$:

$$\begin{aligned} \mathbf{S}_{F_x}(z_s)^{PSVTM} &= \hat{\Phi}(\omega)\left[0, 0, \pm\frac{ik}{\omega^2\bar{\rho}}\frac{F_x}{2}, \pm\left(\frac{\rho_f}{\bar{\rho}} - 1\right)\frac{F_x}{2}, 0, 0, \mp L\frac{F_x}{2}, 0\right]^T \\ \mathbf{S}_{F_x}(z_s)^{SHTE} &= \hat{\Phi}(\omega)\left[0, -i, \left(\frac{\rho_f}{\bar{\rho}} - 1\right)\frac{F_x}{2}, -iL\frac{F_x}{2}, 0\right]^T, \end{aligned} \quad (\text{A11})$$

$$\begin{aligned} \mathbf{S}_{F_y}(z_s)^{PSVTM} &= \hat{\Phi}(\omega)\left[0, 0, \frac{k}{\omega^2\bar{\rho}}\frac{F_y}{2}, -i\left(\frac{\rho_f}{\bar{\rho}} - 1\right)\frac{F_y}{2}, 0, 0, iL\frac{F_y}{2}, 0\right]^T \\ \mathbf{S}_{F_y}(z_s)^{SHTE} &= \hat{\Phi}(\omega)\left[0, \mp\left(\frac{\rho_f}{\bar{\rho}} - 1\right)\frac{F_y}{2}, \mp L\frac{F_y}{2}, 0\right]^T, \end{aligned} \quad (\text{A12})$$

$$\begin{aligned} \mathbf{S}_{F_z}(z_s)^{PSVTM} &= -\hat{\Phi}(\omega)[0, 0, 0, 0, F_z, F_z, 0, 0]^T \\ \mathbf{S}_{F_z}(z_s)^{SHTE} &= [0, 0, 0, 0]. \end{aligned} \quad (\text{A13})$$

A3. EM Sources

[69] The excitation of the porous medium by electric sources is useful to the study of propagation of decoupled EM waves in a fluid-saturated poroelastic stratified medium and to investigate the electro-osmotic coupling effects which create seismic waves from EM waves. *Haartsen and Pride* [1997] have introduced an external current source density $\mathbf{C}$ in their primary set of governing equations, thus allowing the excitation of the medium by EM fields. They derived an expression of the source vector discontinuity for

electric dipoles, which are represented by two electric charges of opposite signs, separated by a distance d .

[70] Let us consider an electric dipole with a moment $m_0 = q_0 d$ located at depth z_s . The current densities associated with dipoles oriented along the x , y , and z directions are written in the form

$$C_i(z_s) = -i\omega m_0 \hat{\Phi}(\omega) \delta(z - z_s), \quad i = x, y, z. \quad (A14)$$

From the general expression of $\mathbf{F}$, and after applying the same transformations as the ones used in the case of an explosion, we obtain the expressions of the source vector discontinuities for the three principal orientations of an electric dipole (C_x , C_y , and C_z):

$$S_{C_x}(z_s)^{PSVTM} = \mp i\omega m_0 \frac{\hat{\Phi}(\omega)}{2} [0, 0, 0, 0, 0, 1, 0] \quad (A15)$$

$$S_{C_x}(z_s)^{SHTE} = \omega m_0 \frac{\hat{\Phi}(\omega)}{2} [0, 0, 1, 0],$$

$$S_{C_y}(z_s)^{PSVTM} = \omega m_0 \frac{\hat{\Phi}(\omega)}{2} [0, 0, 0, 0, 0, 1, 0] \quad (A16)$$

$$S_{C_y}(z_s)^{SHTE} = \mp i\omega m_0 \frac{\hat{\Phi}(\omega)}{2} [0, 0, 1, 0],$$

$$S_{C_z}(z_s)^{PSVTM} = -i\omega d \frac{\hat{\Phi}(\omega)}{\epsilon} \left[0, 0, 0, 0, 0, \tilde{\rho}L, 0, \frac{k}{\omega} \right] \quad (A17)$$

$$S_{C_z}(z_s)^{SHTE} = [0, 0, 0, 0],$$

where $\tilde{\epsilon}$ represents the complex dielectric permittivity [Pride and Haartsen, 1996].

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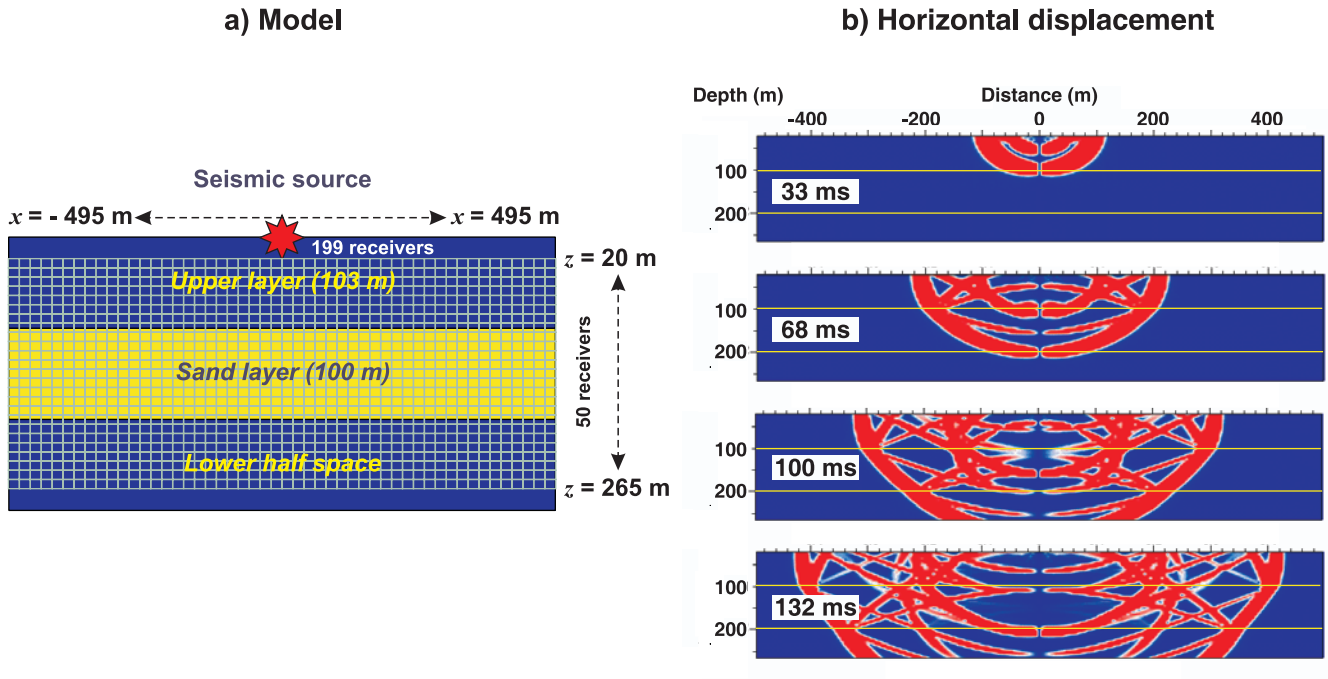


Figure 6. (a) Model used for the computation of snapshots of seismic, electric, and magnetic wave fields. (b) Wave fronts of horizontal displacements generated by a vertical point force at times $t = 32.8$ ms, $t = 68$ ms, $t = 100$ ms, and $t = 132$ ms.

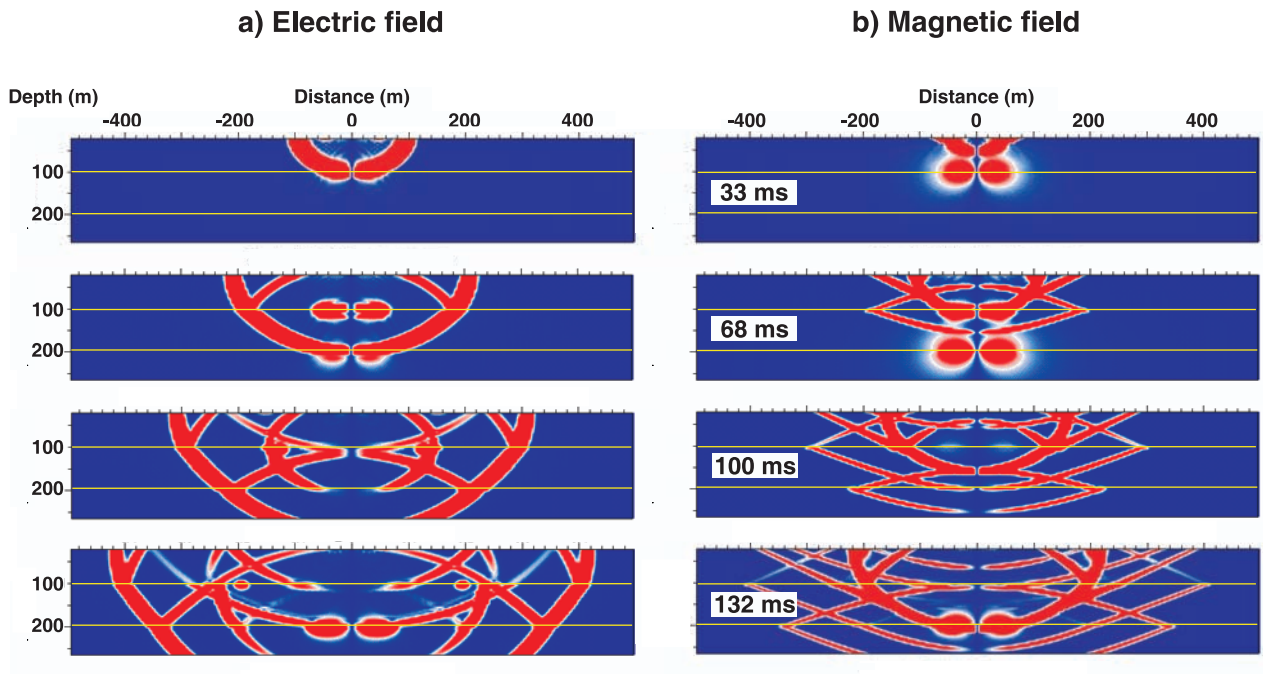


Figure 7. Snapshots of the wave fronts of (a) the electric field E_x and (b) the magnetic field H_y generated by a vertical point force at times $t = 32.8$ ms, $t = 68$ ms, $t = 100$ ms, and $t = 132$ ms.